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Artificial intelligence in polycystic ovary syndrome: a systematic review of diagnostic and predictive applications

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Abstract

Background Polycystic ovary syndrome (PCOS) is one of the most common endocrine disorders, affecting 8–13% of women of reproductive age. Its heterogeneous presentation and the variability of diagnostic criteria make accurate diagnosis and effective management challenging. Artificial intelligence (AI) methods, including machine learning (ML), deep learning (DL), explainable AI (XAI), and large language models (LLMs), have recently emerged as promising approaches to address these gaps.

Objective This systematic review aimed to provide a comprehensive synthesis of AI applications in PCOS, with emphasis on diagnostic performance, biomarker discovery, risk prediction, clinical decision support, model interpretability, and the emerging use of generative AI.

Methods Following PRISMA 2020 guidelines, PubMed, Scopus, and Web of Science were searched from inception to March 2025. Eligible studies applied AI techniques to PCOS and reported at least one performance metric. Two reviewers independently screened and extracted data, with quality appraisal conducted using QUADAS-2 and ROBIS. Given the heterogeneity of designs and outcomes, findings were narratively synthesized across imaging, clinical/EHR, and biomarker/-omics domains.

Results From 662 retrieved records, 80 studies met the inclusion criteria. CNN-based models dominated imaging applications, with accuracies often exceeding 95% and occasionally reaching 98–99%. Supervised ML approaches, particularly random forests and support vector machines, achieved consistent high performance in clinical and biochemical datasets. Omics-based studies revealed novel biomarkers such as HDDC3, SDC2, MAP1LC3A, and OVGP1. However, only about one-quarter of studies applied XAI methods, limiting transparency and clinical trust. Early evaluations of LLMs suggested potential for patient education and decision support but highlighted risks of bias, hallucination, and lack of domain-specific training. Key limitations across studies included small sample sizes, class imbalance, methodological heterogeneity, and limited external validation.

Conclusions AI offers substantial opportunities to advance PCOS diagnosis and prediction by integrating multimodal data and reducing diagnostic subjectivity. Yet its clinical adoption is constrained by interpretability gaps

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and insufficient validation. Future priorities include large multicenter studies, standardized reporting, systematic use of XAI, and careful evaluation of LLMs to ensure safe, equitable, and clinically meaningful integration into PCOS care.

Summary of key findings

AI performance CNN-based imaging models and supervised ML classifiers consistently outperformed traditional diagnostic criteria, reducing subjectivity in ultrasound and clinical assessment.

Biomarker discovery Omics-driven studies identified novel candidate genes (HDDC3, SDC2, MAP1LC3A, OVGP1) with potential for risk stratification.

Explainability Only ~25% of studies applied XAI methods (SHAP, LIME, Grad-CAM). Where used, these improved interpretability and clinician confidence, but most models remained opaque.

Large language models LLMs (ChatGPT, BERT, Gemini) are emerging tools for patient communication, clinical note summarization, and literature synthesis, but raise concerns about hallucination, bias, and lack of domain-specific training.

Risk of bias QUADAS-2 and ROBIS assessments revealed frequent issues with patient selection, dataset representativeness, and lack of external validation. Kaggle datasets were overused, reducing generalizability.

Research priorities Future work should focus on multicenter collaborations, multimodal integration, routine incorporation of XAI, and robust evaluation of LLMs.

Clinical implications AI has the potential to enhance early diagnosis and personalized management of PCOS, but adoption will depend on reproducibility, transparency, and clinician trust rather than accuracy alone.

Keywords Polycystic ovary syndrome, Artificial intelligence, Machine learning, Deep learning, Explainable AI, Large language models, Risk prediction, Biomarkers, Imaging, Clinical decision support

Introduction

Polycystic ovary syndrome (PCOS) is one of the most common metabolic and endocrine disorders in women of reproductive age, with a global prevalence of 8–13% and up to 20% in some regional populations, and has a high burden on public health and the economy. According to the World Health Organization, it is estimated that more than 160 million women worldwide are affected by this syndrome, which indicates the extremely high importance of this disease. Some studies have also shown that in certain regions such as South Asia, the prevalence of PCOS may reach 20%, indicating the geographical and ethnic diversity of this disease [1, 2]. This syndrome is not only influential in terms of reproductive and metabolic problems such as irregular menstruation, infertility, obesity, and insulin resistance, but is also associated with psychological disorders such as depression and anxiety, which significantly reduce the quality of life of patients [3–7]. One of the major challenges in the management of PCOS is its early and accurate diagnosis due to the heterogeneous characteristics of the disease and different diagnostic criteria [3–6]. Beyond its physical burden, PCOS is also associated with depression, anxiety, and impaired quality of life [7].

Despite its prevalence, PCOS is often underdiagnosed or misdiagnosed due to variability in phenotypic expression and inconsistency in applying international diagnostic criteria. Conventional diagnosis combines clinical evaluation, biochemical markers, and ultrasound imaging, but these approaches face persistent challenges: ultrasound interpretation is highly operator-dependent, laboratory thresholds vary across populations, and phenotypic heterogeneity complicates reproducibility [8–11]. Updated international guidelines in 2023 emphasized refined diagnostic criteria and lifestyle modification as a cornerstone of management [12], but translation into practice remains inconsistent.

In this context, the use of artificial intelligence (AI) and machine learning (ML) has been proposed as a new strategy for the prediction, classification and management different diseases specially PCOS. Artificial intelligence models can identify and classify patients with PCOS more accurately and quickly than traditional methods by analyzing complex clinical, hormonal and imaging data. By enabling training models based on large and diverse data, these technologies contribute significantly to the personalization of treatment and prevention of metabolic and psychological complications of the disease. Artificial intelligence can also help reduce the economic burden of the complex treatment and management costs of this syndrome, as it allows for early diagnosis and targeted interventions.

- **Machine learning (ML):** applied for risk stratification, phenotype classification, and biomarker discovery [13–16].
- **Deep learning (DL):** particularly convolutional neural networks (CNNs), which have achieved diagnostic accuracies above 95% in ultrasound and MRI-based PCOS detection [17–20].

- **Explainable AI (XAI):** techniques such as SHAP, LIME, and Grad-CAM that make AI decisions transparent and support clinical trust [21–25].
- **Large language models (LLMs):** transformer-based systems such as GPT and BERT, which can synthesize unstructured text, summarize clinical notes, and support patient communication [26–28].

Recent studies illustrate both the promise and challenges of these methods. CNNs trained on ovarian ultrasound images have reported accuracies approaching 98–99% [17, 18]. Clinical and electronic health record (EHR)-based studies using ensemble methods such as random forests and gradient boosting consistently outperform traditional regression models [14, 15]. Omics-driven studies have identified novel biomarkers such as **HDDC3, SDC2, MAP1LC3A, and OVGP1**, which may refine molecular diagnosis [19, 20]. Yet most studies remain constrained by small sample sizes, class imbalance, methodological heterogeneity, and lack of external validation [29, 30].

Interpretability represents a critical barrier. Only about one-quarter of studies incorporated XAI approaches, despite their ability to reveal which features—such as BMI, AMH, or follicle count—drive model predictions [21–23]. Broader health informatics research has demonstrated the utility of XAI frameworks across diseases: **DeepXplainer** for lung cancer [31], **DiaXplain** for diabetes [32], and fusion-based explainable models for breast cancer [33, 34], and IoMT-driven decision systems [35]. These approaches demonstrate how interpretable pipelines can bridge the "black-box" problem and strengthen clinical adoption, yet PCOS applications remain limited.

Beyond structured data, the rise of generative AI and LLMs offers new opportunities. LLMs such as ChatGPT and BERT can generate clinical summaries, support guideline adherence, and engage in patient education [26–28]. Early evaluations in PCOS contexts suggest potential, but risks of hallucination, bias, and lack of domain-specific training remain significant [36–38]. Governance frameworks are emerging to guide safe use of generative AI [39], and the integration of explainability with LLMs is becoming a new frontier [40].

Another underexplored issue is data source diversity. Most PCOS AI studies use single-center datasets or limited regional samples, raising concerns about generalizability [41]. Recent reviews highlight the importance of dataset transparency, standardized pipelines, and benchmarking practices to improve reproducibility [42, 43].

Objective

To our knowledge, this is the first systematic review that comprehensively synthesizes AI applications in PCOS while explicitly focusing on **explainable AI (XAI)** and the emerging role of **LLMs**. Following PRISMA 2020 guidelines, we critically appraise 80 studies, structured across imaging, clinical/EHR, biomarker/-omics, explainability, and LLM domains. Our aim is to highlight methodological advances, persisting limitations, and future directions for safe, transparent, and clinically meaningful AI integration in PCOS care.

In order to provide an integrated and coherent view, this systematic review has attempted to establish clear links and a coherent flow between these areas related to PCOS by logically integrating and converging different aspects including machine learning (ML), deep learning (DL), explainable artificial intelligence (XAI), and large language models (LLM), avoiding fragmentation and inclusion of disparate elements that reduce readability.

Distinction of the present study

This is the first comprehensive review to simultaneously address two key and relatively new areas of AI research in PCOS: explainable artificial intelligence (XAI) and large language models (LLM). Unlike previous reviews that have focused primarily on the general applications of "artificial intelligence" or "machine learning" in PCOS diagnosis or classification, and have covered each of these areas separately, the present review combines these two perspectives with a particular focus on explainability of models and emerging applications of LLM in medicine and clinical data analysis. Specifically, this work, in addition to reviewing conventional imaging models and classification algorithms, deeply explores the impact of XAI in helping to increase clinical confidence and decision-making transparency, and the role of LLMs in the generation of clinical texts, knowledge extraction, and communication with patients. Therefore, this combination of topics, while addressing gaps in previous reviews, provides new guidance for research and clinical implementation of AI in PCOS, specifically addressing emerging challenges and opportunities in these two areas.

Materials and methods

Reporting framework

This systematic review was designed and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [44]. Because many of the included studies were predictive modeling investigations, we also referred to the CONSORT-AI [45], SPIRIT-AI [46], and DECIDE-AI [47] guidelines to ensure contextual best practices. In addition, we consulted the TRIPOD+AI extension [48] to evaluate reporting quality in prediction studies. These frameworks were not directly applicable to our dataset (which did not include randomized trials or early-phase interventional studies) but provided context for aligning PCOS AI research with emerging reporting standards.

Eligibility criteria

We included peer-reviewed primary research articles that applied artificial intelligence (AI) techniques to the diagnosis, prediction, or management of PCOS, published in English between January 2000 and 2 March 2025.

Definition of AI application. For this review, an "AI application" was defined as a computational approach capable of learning from data and producing predictions, classifications, or explanations beyond conventional statistical analysis. Eligible AI methods included:

- **Supervised learning:** support vector machines, random forests, decision trees, k-nearest neighbors, naïve Bayes, and logistic regression when optimized in ML pipelines.
- **Deep learning:** convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders, and transformer-based models.
- **Unsupervised and semi-supervised learning:** clustering and dimensionality reduction methods (e.g., k-means, PCA, t-SNE, UMAP).
- **Explainable AI (XAI):** interpretability methods such as SHAP, LIME, Grad-CAM, saliency mapping, attention visualization, DALEX, and QLattice.
- **Large language models (LLMs):** transformer-based systems (e.g., GPT, BERT, BioBERT) when applied to PCOS-related literature synthesis, clinical notes, or electronic health records.

Inclusion criteria: Studies had to report at least one quantitative performance metric (e.g., accuracy, sensitivity, specificity, AUC, precision, recall, F1-score).

Exclusion criteria: We excluded (1) studies relying only on conventional statistics without AI components; (2) papers lacking methodological or performance detail; (3) reviews, editorials, abstracts, or case reports without full text; and (4) duplicates; (5) Conference Paper.

Information sources and search strategy

A comprehensive literature search was conducted in **PubMed, Scopus, and Web of Science** from database inception to 2 March 2025. Boolean operators (AND, OR) combined AI-related terms ("artificial intelligence," "machine learning," "deep learning," "explainable artificial intelligence," "large language model," "LLM") with PCOS-related terms ("polycystic ovary syndrome," "PCOS," "Stein-Leventhal syndrome").

The full search strings for each database, with field tags and syntax, are provided in Supplementary Table S1. Reference lists of included papers and relevant reviews were also hand-searched to capture additional eligible studies.

Study selection

All retrieved records were imported into a reference manager. Duplicates and non-eligible record types (e.g., conference papers) were removed. Two reviewers independently screened titles and abstracts, followed by full-text assessment of potentially relevant studies against eligibility criteria. Discrepancies were resolved by consensus or adjudicated by a third reviewer. A PRISMA flow diagram (Fig. 1) details the number of studies retrieved, screened, excluded, and included.

Data extraction

A standardized extraction template was piloted and refined. Two reviewers independently extracted:

- Bibliographic details (authors, year, country).
- Study design (prospective, retrospective, cross-sectional, case–control, or trial).
- Dataset characteristics (sample size, modality, origin, public vs proprietary).
- Input features (imaging, clinical, biochemical, genetic, multimodal).
- AI methods (algorithm type, preprocessing, validation strategy).
- Outcome measures (accuracy, AUC, sensitivity, specificity, F1-score, precision, recall).
- XAI techniques, if reported.
- Main findings and stated limitations.

Quality assessment and risk of bias

Two reviewers independently assessed study quality. Tools applied:

- **QUADAS-2** The Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool, adapted for AI-based studies [49], assessed risk of bias and applicability across four domains: Patient Selection/Eligibility, Index Test, Reference Standard, Flow & Timing/Synthesis Applicability Concerns. Two reviewers independently categorized risk as Low, Unclear, or High, with inter-rater agreement quantified using Cohen's kappa. Results are in Supplementary Table S2a.
- **ROBIS** for review-type studies. Results are in Supplementary Table S2b.

Disagreements were resolved by consensus. Full domain-level ratings are presented in Supplementary Table S2.

Data synthesis

Given heterogeneity across study designs, datasets, and outcomes, we performed a narrative synthesis instead of meta-analysis. Results were grouped thematically into:

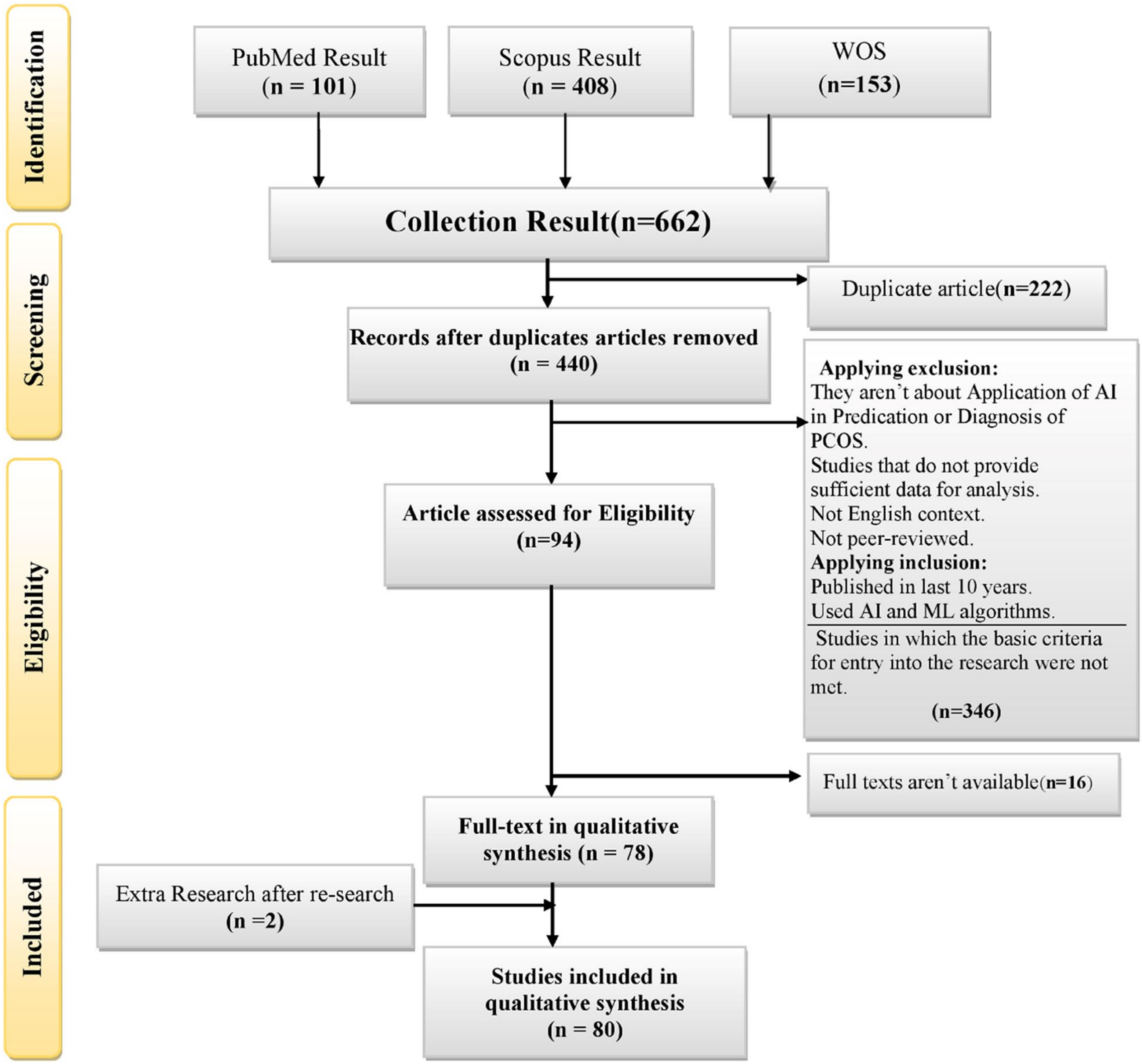

Fig. 1 Prisma flowchart of search and exclusion process. Out of 662 retrieved studies, 80 studies were selected for data analysis

- Imaging-based studies (Table 2).
- Clinical/EHR-based studies (Table 3).
- Omics/biomarker studies (Table 4).

Results

Study selection

From 662 records initially identified (SCOPUS: 408, Web of Science: 153, PubMed: 101), 245 unique studies remained after removing duplicates and conference abstracts. Following title and abstract screening, 116 studies underwent full-text review. Ultimately, 80 studies met the inclusion criteria. The selection process is illustrated in the PRISMA flow diagram (Fig. 1).

Study characteristics

The 80 included studies spanned diverse geographical regions, with the largest contributions from India ($n = 19$), China ($n = 15$), and the United States ($n = 12$). Study designs comprised retrospective cohorts ($n = 25$), cross-sectional studies ($n = 18$), case–control studies ($n = 14$), and randomized controlled trials ($n = 9$). Sample sizes varied considerably, ranging from fewer than 50 participants to more than 30,000 records. The geographic and study design distributions are summarized in Fig. 2.

Figure 2 shows the geographical distribution and study designs of the included studies, indicating that the most research activity in the field of AI applications in PCOS is from countries such as India, China and the United States. Also, the dispersion of study design types includes

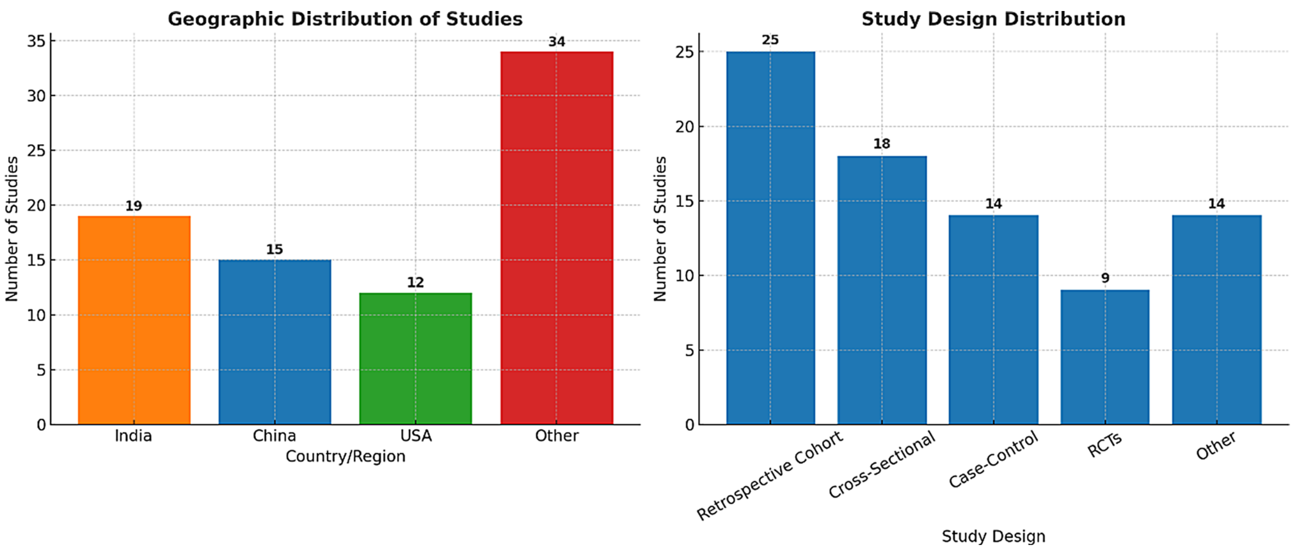

Fig. 2 Geographic and study design distribution of included studies on AI applications in PCOS

retrospective cohorts, cross-sectional studies, case-control studies and randomized controlled trials. The observed differences in geographical distribution and study design could be due to several factors such as access to data sources, research infrastructure, national research priorities and health policies. This diversity clearly demonstrates the importance of multicenter collaborations and standardization of methods to strengthen the validity and generalizability of AI research results in PCOS.

Overview of AI applications

This section provides a comprehensive summary of the reviewed studies, reflecting the broad scope of research conducted on the application of artificial intelligence (AI) in polycystic ovary syndrome (PCOS). Key information from each study, including the type of data used, sample size, subject area (imaging, clinical data/electronic health records, and molecular/biochemical data), the main AI algorithms employed, and the performance measures reported, are presented in an organized and categorized manner. This structure allows the reader to have an overview of the trends, diversity of methods, and quality of data in this area, and provides a prelude to more detailed analyses in the tables and sections that follow.

To provide a comprehensive overview of the different areas of application of artificial intelligence (AI) in polycystic ovary syndrome (PCOS), all eligible studies for this systematic review are summarized and summarized in Table 1. The table displays key information for each study, including the year of publication, country of study, subject area (imaging, clinical data and electronic health records, or biomolecular data), volume and nature of data used, and the main AI methods used.

This overview allows for a quick and transparent comparison of the different approaches, sample sizes, and analytical methods, and helps identify the dominant research trends in each subfield. This comprehensive review also provides a basis for more specialized analysis in subsequent tables, which are dedicated to each area in detail and describe the performance outcomes and key achievements of each category. Such an organization of information provides a coherent framework for understanding the progress, strengths, and limitations of current research in artificial intelligence for PCOS. Table 1 provides a snapshot overview of the 80 studies included in this review, summarizing their geographic origin, study domain, dataset size, and main AI approaches. This snapshot highlights the diversity of methods applied—from machine learning on small clinical cohorts to deep learning on large imaging datasets—and illustrates how research activity has been distributed across countries and methodological domains. To improve clarity, studies are thematically grouped into three domains:

- Table 2. Imaging-based AI studies (ultrasound, MRI, segmentation, follicle counting).
- Table 3. Clinical and EHR-based AI studies (structured health records, survey data, multimodal features).
- Table 4. –Omics and biomarker-based AI studies (genomic, transcriptomic, proteomic, metabolomic, lipidomic datasets).

The detailed extraction of article information based on AI algorithms and methodological features, previously included as Table 1, has been moved to the Supplementary section (Supplementary Table S3).

Table 1 Overview of included studies on AI applications in PCOS

Author(s), Year [Ref]	Domain	Sample size/Dataset	Main AI methods
Zhang et al., 2018, China [50],	Biomarker/-omics	306 PCOS vs 13,681 non-PCOS genes	SVM
Ho et al., 2020, Taiwan [51],	Biomarker/-omics	48 PCOS, 181 controls	SVM
Cheng et al., 2019, USA [8],	Clinical/EHR (text)	39,093 US reports	RBC, GBT
Thakre et al., 2020, India [10],	Clinical	Kaggle PCOS dataset (~541)	RF, SVM, LR, NB, KNN
Xie et al., 2020, China [9],	Biomarker/-omics	76 PCOS, 57 controls	RF, ANN
Vikas et al., 2021, India [52],	Imaging (ultrasound)	200 images (100 PCOS, 100 non)	CNN, VGG-16
Kodipalli et al., 2021, India [53],	Clinical (survey + clin.)	624 analyzed	SVM, DT, KNN, Fuzzy TOPSIS
Mehr et al., 2022, Turkey [54],	Clinical	541 (Kaggle)	RF, ExtraTrees, AdaBoost, MLP
Huang et al., 2021, China [55],	Biomarker (Raman spectra)	300 (150 PCOS, 150 non)	ANN, PCA
Kiran et al., 2022, India [56],	Clinical	1600 hospital records	BKIP classifier
Kiran et al., 2022, India [57],	Clinical	1800 hospital records	Hybrid SVM+LR
Lv et al., 2022, China [16],	Imaging (scleral)	721 subjects	U-Net, ResNet-18
Agrawal et al., 2022, India [58]	Review	—	CNN, SVM (review)
Zigarelli et al., 2022, USA [59],	Clinical (self-screen)	526 (170 PCOS, 356 non)	CatBoost
Sreejith et al., 2022, India [12],	Clinical (CDSS)	541	RF + RDA
Suha et al., 2022, Bangladesh [60],	Imaging (US)	594 images	VGG-16 + XGBoost
Na et al., 2022, China [14],	Biomarker/-omics	Multi-GEO datasets	LASSO, SVM-RFE
Nasim et al., 2022, Pakistan [61],	Clinical	541	GNB, RF, SVM, GBC
Kavitha et al., 2023 [20]	Clinical	541	CatBoost, XGB, LGBM
Yeruva et al., 2023, India [62],	Clinical + Imaging	541 clinical; 53 US images	RF, CNN
Raghav et al., 2023, India [63]	Clinical	543	RF, DT, SVM, LR
Swapnarekha et al., 2023, India [64]	Clinical	541 (Kaggle)	Extreme Learning Machine (Bayesian opt.)
Yamini et al., 2023, India [65],	Clinical	541 (Kaggle)	RF, LR, SVM, NB, KNN, XGBoost
Lim et al., 2023, China [66],	Clinical (TCM + pulse)	450 women	SVM, RF, XGB, ET
Alagarsamy et al., 2023, India [67],	Clinical	541 (Kaggle)	SVM, KNN, NB, Ensemble
Abouhawwash et al., 2023, Egypt/USA/India/Saudi Arabia [68]	Clinical	541 (Kaggle)	CNN, MLP, RNN, Bi-LSTM
Nsugbe, 2023,UK [15]	Clinical	541 (Kaggle)	DT, LDA, LR, KNN, SVM
Batra et al., 2023, India [69]	Clinical	541 (Kaggle)	RF, SVM, LR
Pandey et al., 2023, India [70]	Clinical	541 (Kaggle)	CatBoost, RF, KNN, SVM
Sumathi et al., 2024, India [17]	Imaging (US)	1918 images	DarkNet-19, AlexNet, SqueezeNet, SVM
Khanna et al., 2023, India [71]	Clinical	541 (Kaggle)	RF, XGB, AdaBoost, LR, DNN, 1D-CNN
Alamoudi et al., 2023, Saudi Arabia [72]	Imaging + Clinical	391 US images + 285 clinical	InceptionV3, VGG, MobileNet, Fusion
Chen et al., 2023, China [73]	Biomarker/-omics	PCOS + RIF datasets (GEO)	LASSO, SVM-RFE, RF
Yang et al., 2023, China [74]	Imaging (MRI)	20 women (40 ovaries)	DL reconstruction (AIRTM Recon DL)
Lim et al., 2023, China [75]	Clinical (pulse)	459 subjects	KNN, SVM, RF, LSTM
Sheikdavood et al., 2023, India [76]	Imaging (US segmentation)	—	CNN, DNN, Adaptive K-means, RSA
Elmannai et al., 2023 [77]	Clinical	541 (Kaggle)	LR, RF, DT, NB, SVM, KNN, XGB, Stacking
Wu et al., 2023, China + international [78]	Biomarker/-omics	1000 infertile women	LR, SVM, KNN, RF, GBDT, NN
Moral et al., 2024, India [21]	Clinical	—	PODBoost (hybrid boosting)
Gunesli et al., 2024, Turkey [24]	LLMs (ChatGPT, Gemini)	—(AI responses to PCOS Qs)	LLM evaluation
Paramasivam et al., 2024, India [79]	Imaging (US)	277 images	Self-defined CNN, RF, SVM, LR
Rona et al., 2024, Turkey [80]	Imaging (MRI radiomics)	202 ovaries (101 women)	RF, GBC, blend model
Galagan et al., 2024, Ukraine [81]	Imaging (US)	3,846 images	CNN
Praneesh et al., 2024, India/Malaysia/Thailand [82]	Imaging (US)	—	VGG19, RF, LR, SVM, ANN
Bedi et al., 2024, India/Malaysia/UAE/Jordan [83]	Imaging (US)	1,639 images	Attention Residual U-Net (AResUNet)

Table 1 (continued)

Author(s), Year [Ref]	Domain	Sample size/Dataset	Main AI methods
Rao et al., 2024, India [84]	Clinical	541 (Kerala hospitals)	Optimized DL (NN, Optuna+GA), SVC, RF
Rahman et al., 2024, Bangladesh [85]	Clinical	541 (Kaggle)	LR, DT, AdaBoost, RF, SVM
Wu et al., 2024, China [28]	Biomarker/-omics (proteomics)	33 PCOS, 7 controls	SVM, RF, XGB, GLM, LASSO
Bairagi et al., 2024, India/Thailand/Philippines [86]	Imaging (US)	1924 images	CNN
Devranoglu et al., 2024, Turkey [25]	LLMs (ChatGPT-4)	460 queries	LLM evaluation
Emanuel et al., 2024, NZ/Germany [26]	Clinical (lab test clustering)	1,585 test sets (Reddit)	K-means, KNN
Farhan et al., 2024, Bangladesh [87]	Clinical	541 (Kaggle)	LR, NB, AdaBoost, RF, ANN
Umaa Mahesswari et al., 2024, India [23]	Clinical (XAI)	541 (Kaggle)	RF, LR, SVM, DT, NB, KNN, AdaBoost, XGB
Gülhan et al., 2024, Turkey [22]	Imaging (US)	54 images (augmented to 270)	CNN, TL (SqueezeNet)
Sai et al., 2024, India/South Korea [88]	Imaging + Clinical	1468 US images + clinical data	ResNet-50, SVM, RF, LR
Narni et al., 2024, India [89]	Clinical (screening tool)	200 women (100 PCOS, 100 controls)	RF, Feedforward NN
Ahmad et al., 2024, India [90]	Clinical	540 samples (balanced)	CNN, LSTM, CNN-LSTM hybrid
Zad et al., 2024,USA [27]	Clinical/EHR	30,601 women (BMC)	LR, SVM, GBT, RF, MLP
Wang et al., 2024,China [19]	Biomarker/-omics	70 PCOS, 60 controls	XGBoost, AdaBoost, DT, KNN, LR, RF
Yadav et al., 2024,India [91]	Clinical	512 (SMOTE balanced)	GB, XGB, AdaBoost, RF
Selvaraj et al., 2024, India [92]	Social media (Reddit)	24,794 posts	LDA, SSG_LLDA, Apriori ARM
Yang et al., 2024,China [18]	Biomarker/-omics	32 granulosa + 12 oocyte samples	LASSO, RF, SVM-RFE
Mridul et al., 2024, Bangladesh/USA [93]	Clinical (Kaggle)	540	ANN, RNN, CNN, LSTM, RF, SVM, etc.
Sowmiya et al., 2024,India [94]	Imaging (US)	100–200 images (private SRM hospital)	YOLOv8, F-Net, MobileNet, ResNet, RF
Poorani et al., 2024,India [95]	Imaging (US segmentation)	1928 train, 1552 test	MOT-SF (novel segmentation)
Kumar et al., 2024, India [96]	Clinical	541 (Kaggle)	LR, LDA, GNB, SVM, XGB, RF, KNN, AdaBoost
Moral et al., 2024, India [21]	Imaging (US)	1,924 train, 1,932 test	CystNet (InceptionNet + Autoencoder)
Kermanshahchi et al., 2024,USA [97]	Imaging (US)	1,932 images	CNN
Shanmugavadivel et al., 2024, India/Saudi Arabia/Ethiopia [98]	Imaging + Clinical	541 clin., 3856 US images	SVM, CNN, VGG16 (transfer)
Wang et al., 2024, China [99]	Biomarker/-omics	29 women (two GEO sets)	LASSO, SVM-RFE
Tan et al., 2024, China [100]	Biomarker/-omics	22 PCOS, 14 controls	RF, SVM, DT, KNN
Chen et al., 2025, China [101]	Biomarker/-omics (lipidomics)	152 PCOS, 50 controls	LR, RF, SVM (ensemble)
Mohiuddin et al., 2025, Bangladesh/Australia [102]	Clinical	541 (Kaggle)	LR, SVM, RF, KNN, AdaBoost, CatBoost
Yang et al., 2025,China [38]	Imaging (MRI)	22 women (44 ovaries)	CNN (DL MRI reconstruction)
Rona et al., 2025,Turkey [103]	Imaging (MRI radiomics)	101 PCOS, 40 controls	LightGBM, RF, Lasso
Panjwani et al., 2025,India [104]	Clinical (non-invasive)	932 blended samples	WaOEL (stacked ensemble), DL, RF
Parvathi et al., 2025,India [105]	Clinical	541 (Kaggle)	DNN, CNN, RNN
Chen et al., 2025,China [106]	Biomarker/-omics	13 controls, 25 PCOS	SVM, XGBoost
Wang et al., 2025, Taiwan [107]	Clinical (MJ Health DB)	1120 (170 PCOS, 950 controls)	RF, SGB, MARS, XGB, CatBoost
Suha & Islam, 2023, Bangladesh [108]	Clinical	541 (Kaggle)	Stacking ensemble (SVM, LR, DT, KNN, NB + GB)

Imaging-based AI studies

Studies related to the application of artificial intelligence in medical imaging of polycystic ovary syndrome (PCOS) constitute an important part of the research in this field. These studies have enabled more accurate diagnosis and quantification of key features of the disease by analyzing images from ultrasound, MRI, and other modalities. Table 2 provides a summary of the main characteristics and methods used in these imaging studies to clearly visualize the research trends and technologies used.

Table 2 Imaging-based AI studies in PCOS ($n=23$)

Author, Year [Ref]	Modality	No (images/subjects)	Main model(s)	XAI	Best reported performance	External validation/ Notes
Vikas B., 2021 [52]	US	200 imgs	CNN (VGG-16 TL)	—	Acc 98%	Train/val/test split; private data.
Lv W., 2022 [16]	Scleral	721 subj	U-Net+ResNet-18+MIL	Grad-CAM	AUC 0.979, Acc 92.9%	5-fold CV; non-ovarian, non-invasive screen.
Suha S.A., 2022 [60]	US	594 imgs	VGG16 features + stacking (XGBoost meta)	—	Acc 99.89%	70/30 split; mixed public/ clinical; no external set.
Yeruva S., 2023 [62]	US (+clinical)	53 imgs (+541 records)	RF (clinical), CNN (US)	—	CNN Acc 93%	App prototype; small US set.
Sumathi M., 2024 [17]	US	1,918 imgs (+25 real-time)	DarkNet-19, AlexNet, SqueezeNet, SVM	—	DarkNet-19 Acc 99%, Sens 100%, F1 98%	Mostly Kaggle; limited real-world images.
Alamoudi A., 2023 [72]	US ± clinical	391 imgs; 285 clinical	InceptionV3, VGGs, Mo-bileNet; fusion	—	InceptionV3 Acc 84.8% (fusion 82.5%)	Single-center; 10-fold CV.
Yang R., 2023 [74]	MRI	40 ovaries (20 pts)	DL recon (AIR™ Recon DL)	—	ICC↑ (SSFSE-DL 0.93)	Imaging quality/repeat-ability study, not classifier.
Sheikdavood K., 2023 [76]	US (segm.)	Not stated	CNN/DNN + Improved k-means + RSA	—	Reported ↑segm. vs baselines	Metrics limited in paper.
Paramasivam G.B., 2024 [79]	US	277 imgs	SD-CNN + RF/SVM/LR	—	Acc 96.43% (SD-CNN+RF)	Mixed private/Kaggle; grid search.
Rona G., 2024 [80]	MRI radiomics	202 ovaries	RF, GBC, blend	—	AUC 0.70; Acc 73%	Distinguish classical vs non-classical PCOS.
Galagan R., 2024 [81]	US	3,846 imgs	CNN	Grad-CAM	Acc 99.7%	Kaggle; generalizability concern noted.
Praneesh M., 2024 [82]	US	—	VGG19, RF, LR, SVM, ANN	—	VGG19 Acc 96%	Multi-model comparison; dataset not detailed.
Bedi P., 2024 [83]	US (segm.)	1,639 imgs (294 pts)	Attention Residual U-Net	—	Acc 98%, mIoU 0.99	MMOTU dataset; 70/30 split.
Bairagi V.K., 2024 [86]	US	1,924 train/1,468 test	CNN	—	Acc 97.76%, Sens 99.27%	Kaggle; no external cohort.
Gülhan P.G., 2024 [22]	US (follicle detect.)	54 imgs (aug to 270)	CNN TL (SqueezeNet); Wiener+thr.	—	Acc 77.8% (segmented)	Very small set; segmenta-tion best 97.6% detect.
Nethra Sai M., 2024 [88]	US (+clinical)	3,200 train/1,468 test	ResNet-50; SVM/RF/LR	SHAP	SVM Acc 99%	Combines clinical fea-tures; SHAP for features.
Sowmiya S., 2024 [94]	US	100–200 imgs	YOLOv8 (detect) + F-Net (CNN)	—	Acc 95–97.5%; AUC 0.99	GAN augmentation; private SRM data.
Poorani B., 2024 [95]	US (segm.)	1,928/1,552	MOT-SF (novel Otsu fusion)	—	Acc 85.1%; Jac-card 74.2%	Segmentation focus; parameter-sensitive.
Moral P., 2024 [21]	US	1,924/1,932	**CystNet** (InceptionV3+Autoenc) + RF	—	Acc 97.75%; Sens 98.29%	5-fold CV; compute-heavy.
Kermanshah-chi J., 2024 [97]	US	1,932 imgs	CNN	—	Acc 100%; AUC 1.00; F1 0.905	Possible overfit; precision 82.6%.
Shanmugava-divel K., 2024 [98]	US + clinical	3,856 US; 541 clinical	CNN, **VGG16 TL**, SVM	—	VGG16 Val Acc 98.29%; SVM 94.44%	Multi-modality; suggests attention mech.
Yang R., 2025 [38]	MRI	22 pts (44 ovaries)	DL recon (high-res)	—	Repeatability ↑ (ICC); clarity ↑	Not a diagnostic classifier.
Rona G., 2025 [103]	MRI radiomics	101 PCOS, 40 ctrls	**LightGBM** (best)	—	Acc 82%; AUC 0.80; F1 0.88	Retrospective; manual segmentation.

Abbreviations: US ultrasound; TL transfer learning; segm. segmentation; ICC intraclass correlation coefficient

Table 3 Clinical and EHR-based AI studies in PCOS ($n=36$)

Author, Year, Country [Ref]	Dataset	Main AI models	XAI	Key findings
Cheng, 2019,USA [8]	39,093 EMR US reports	RBC, GBT	—	PCOM classification Acc ~97%
Thakre, 2020, India [10]	541 (Kaggle)	RF, SVM, LR, NB, KNN	—	RF best, Acc ~91%
Kodipalli, 2021,India [53]	624 surveys + clinical	SVM, DT, KNN, Fuzzy TOPSIS	—	Fuzzy TOPSIS Acc 98.2%
Mehr, 2022,Turkey [54]	541 (Kaggle)	RF, ExtraTrees, AdaBoost, MLP	—	RF + feature selection Acc 98.9%
Kiran, 2022,India [56]	1600 records	BKIP classifier	—	Acc 89%
Kiran, 2022,India [57]	1800 records	Hybrid SVM+LR	—	Acc 89%, AUC 1.0
Zigarelli, 2022,USA [59]	526 (170 PCOS)	CatBoost	SHAP	Acc ~90%
Sreejith, 2022,India [12]	541	RF + RDA	—	Acc ~90%
Nasim, 2022,Pakistan [61]	541	GNB, RF, SVM, GBC	—	GNB Acc 100%
Kavitha, 2023,India [20]	541	CatBoost, XGB, LGBM	—	CatBoost best (Acc 94%)
Raghav, 2023,India [63]	543	RF, DT, SVC, LR	—	RF Acc 92.7%
Swapnarekha, 2023,India [64]	541 (Kaggle)	ELM (Bayesian opt.)	—	Acc 99.3%
Yamini, 2023,India [65]	541	RF, LR, SVM, NB, KNN, XGB	—	RF best (Acc 90%)
Lim, 2023, China [66]	450 women	SVM, RF, XGB, ET	SHAP	SVM AUC 0.878
Alagarsamy, 2023,India [67]	541	SVM, KNN, NB, Ensemble	—	Ensemble Acc 97.2%
Abouhawwash,2023,Egypt [68]	541	CNN, MLP, RNN, Bi-LSTM	—	CNN Acc 98.7%
Nsugbe, 2023,UK [15]	541	DT, LDA, LR, KNN, SVM	—	SVM best
Batra,2023,India [69]	541	RF, SVM, LR	—	RF Acc 92%
Pandey,2023, India [70]	541	CatBoost, RF, KNN, SVM	—	CatBoost Acc 97%
Khanna, 2023, India [71]	541	RF, XGB, AdaBoost, LR, DNN, CNN	SHAP, LIME	STACK-3 Acc 98%
Rao, 2024, India [84]	541	Opt. DL (NN, Optuna+GA), SVC, RF	—	DL Acc 93.6%
Rahman, 2024 [85]	541	LR, DT, AdaBoost, RF, SVM	—	RF & AdaBoost Acc 94%
Farhan, 2024, Bangladesh [87]	541	LR, NB, AdaBoost, RF, ANN	—	LR Acc 95%
Umaa Mahesswari, 2024, India [23]	541	RF, LR, SVM, DT, NB, KNN, AdaBoost, XGB	SHAP, LIME	RF Acc 99.3%
Narni, 2024, India [89]	200	RF, NN	—	RF Acc 95% train
Ahmad, 2024, India [90]	540	CNN, LSTM, CNN-LSTM	SHAP, LIME	CNN Acc 96.6%
Zad, 2024,USA [27]	30,601 EHR	LR, SVM, GBT, RF, MLP	—	AUC 0.85
Yadav, 2024, India [91]	512 (SMOTE)	GB, XGB, AdaBoost, RF	—	Acc 94%
Selvaraj, 2024, India [92]	24,794 Reddit posts	LDA, SSG_LLDA, ARM	—	Identified symptom subtypes
Mridul, 2024, Bangladesh/USA [93]	540	ANN, RNN, CNN, LSTM, RF, SVM, etc.	—	Boosted RF & SVC Acc 99.3%
Kumar, 2024, India [96]	541	LR, LDA, GNB, SVM, XGB, RF, KNN, AdaBoost	—	LR+PSO Acc 96.3%
Shanmugavadivel, 2024, India [98]	541 + 3856 US	SVM, CNN, VGG16	—	VGG16 Acc 98.3%
Mohiuddin, 2025, Bangladesh/Australia [102]	541	LR, SVM, RF, KNN, AdaBoost, CatBoost	—	Acc 99.8%
Panjwani, 2025, India [104]	932 blended	WaOEL, DL, RF, SVM	SHAP	WaOEL AUC 0.93
Parvathi, 2025, India [105]	541	DNN, CNN, RNN	—	DNN Acc 81%
Wang, 2025, Taiwan [107]	1120 (170 PCOS)	RF, SGB, MARS, XGB, CatBoost	—	MARS Acc 79%

Table 2 summarizes studies that applied AI to clinical and electronic health record (EHR) data. These works often used structured features such as demographics, hormone levels, and metabolic indicators, with ensemble learning and traditional machine learning methods (e.g., Random Forest, SVM) being the most common. While many achieved high accuracy, most relied on regional or Kaggle-based datasets, underscoring the need for larger, multi-center validation.

The pie chart below (Fig. 3) shows the distribution of medical imaging modalities used in AI studies related to the diagnosis and analysis of polycystic ovary syndrome (PCOS). As can be seen, the use of ultrasound images is the most frequent, with a share of approximately 67%, reflecting the importance and widespread use of this non-invasive imaging modality in the diagnosis of PCOS. This is followed by MRI images, with a share of approximately 19%, playing an important role in advanced image analysis and radiomics. Also, studies focused on ultrasound image segmentation account for about 11% of the total studies, indicating a focus on more accurate feature extraction from images.

In addition, the more limited use of scleral images and specific follicle detection techniques, with shares of less than 4%, indicate the wide variety of imaging data that,

Table 4 Omics and biomarker-focused AI studies in PCOS ($n=16$)

Author, Year, Country [Ref]	Data type & size	Main AI models	XAI	Key biomarkers/signals & main results
Zhang, 2018,China [50]	Gene networks (13,681 genes; 306 PCOS labels)	SVM	Feature ranking	Predicted 233 candidate PCOS genes; AUC ≈0.80.
Ho, 2020, Taiwan [51]	Ovarian gene expression (48 PCOS, 181 ctrls)	SVM (functionome)	—	Dysregulated immune/metabolic pathways; AUC ≈0.99 (CV).
Xie, 2020,China [9]	GEO transcriptomics (multi-set)	RF, ANN ("neuralPCOS")	—	12-gene panel; AUC 0.73 (microarray), 0.65 (RNA-seq).
Huang, 2021,China [55]	Raman spectra of follicular fluid (150 PCOS, 150 ctrl)	ANN	—	Oocyte competence prediction Acc 90%; pregnancy prediction Acc 74%.
Na, 2022,China [14]	GEO microarray/RNA-seq (5 datasets)	LASSO, SVM-RFE	—	Biomarkers HDDC3, SDC2; AUC 0.92 and 0.82; immune infiltration shift.
Chen W, 2023, China [73]	GEO (PCOS & RIF cohorts)	WGCNA, LASSO, SVM-RFE, RF	—	Shared genes GLIPR1, MAMLD1; AUCs ~0.72–0.88 across tasks.
Wu X., 2023, China [78]	Multi-omic clinical/genetic ($n=1000$)	LR, SVM, RF, GBDT, NN	—	Genetic & metabolic signatures predict ovulation response; external validation.
Wu Y.Y., 2024, China [28]	Proteomics (PCOS pregnancy prognosis)	SVM, RF, GLM, XGB, LASSO	—	Signature predicted pregnancy loss; AUC ≈0.94 (bootstrap).
Emanuel (RHK), 2024, NZ/Germany [26]	Community lab panels ($n=1,585$ sets, Reddit)	K-means (+KNN impute)	—	Four to seven biochemical phenotypes (metabolic vs reproductive clusters).
Wang F.J, 2024, China [19]	Clinical + OVGP1 protein (70 PCOS, 60 ctrls)	XGBoost, AdaBoost, DT, KNN, LR, RF	—	OVGP1 + 6 markers improved diagnosis; best AUC 0.995 (XGB).
Yang Y.Z., 2024, China [18]	GC & oocyte transcriptomes (GEO)	LASSO, RF, SVM-RFE	—	MAP1LC3A (mitophagy) diagnostic AUROC 0.86–0.96; immune shifts.
Wang J.Q., 2024 [99]	GEO (GSE6798, GSE48301)	LASSO, SVM-RFE	SHAP	15 acetylation-related genes; AUC 0.962 (train), 0.924 (valid).
Tan C.Y., 2024, China [100]	Multi-dataset transcriptomics	RF, SVM; WGCNA	—	Oxidative stress–lipid metabolism clusters; five hub genes (AUC up to 0.96).
Chen J.Y., 2025, China [101]	Lipidomics (152 PCOS, 50 ctrls)	LR, RF, SVM (ensemble screening)	—	Serum lipid species PI(18:0/20:3)-H, PE(18:1p/22:6)-H; AUC 0.815.
Chen W.X., 2025, China [106]	RNA-seq (own + GEO)	SVM, XGBoost; LASSO, SVM-RFE	SHAP	CNTN2, CASR, CACNB3, MFAP2; AUC 0.80–0.88; reduced CD4 memory T cells.
Wang C.Y., 2025, Taiwan [107]	Health-check biomarker panel ($n=1120$)	RF, SGB, MARS, XGB, CatBoost	Feature importance	Key risks: age, TG, GPT, WBC, UA, platelets; best Acc 79% (MARS).

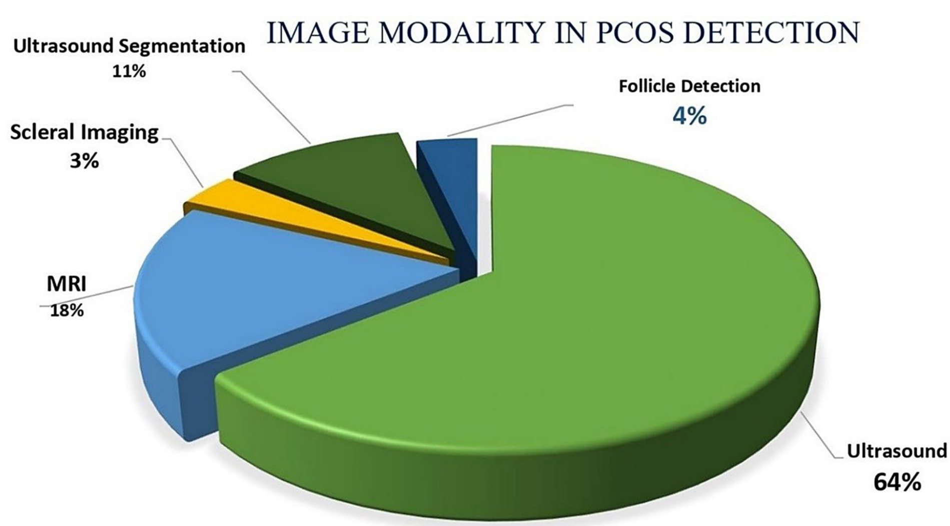

Fig. 3 Distribution of medical imaging types used in AI studies for PCOS diagnosis

despite the limited number of studies, are recognized as emerging and potential areas for AI research in PCOS.

Clinical and EHR-based AI studies

To shed light on the role of AI in the analysis of clinical data related to polycystic ovary syndrome (PCOS), a series of selected studies that used structured clinical data, including biochemical tests, clinical indices, and electronic health record (EHR) data, were reviewed. These studies have provided diverse solutions for the diagnosis, prediction, and management of PCOS using various machine learning algorithms. Table 3 presents key characteristics of these studies, including sample size, data type, AI models used, and key performance results, to provide a comprehensive picture of research trends and challenges in the field of clinical data.

Table 3 presents studies that focused on –omics and biomarker discovery in PCOS, spanning genomics, transcriptomics, proteomics, metabolomics, and lipidomics. These investigations applied machine learning to uncover novel molecular signatures and pathways, identifying candidate biomarkers such as HDDC3, SDC2, MAP1LC3A, and OVGP1. While these findings shed light on disease mechanisms and potential diagnostic tools, most studies were limited by small sample sizes and require experimental validation.

Omics and biomarker studies

AI studies in the field of biomarker and omics data analysis in polycystic ovary syndrome (PCOS) constitute an important part of molecular research. These studies have contributed to the identification of novel biomarkers and more accurate prediction models by analyzing genomic, transcriptomic, proteomic, metabolomic, and lipidomic data. Table 4 provides a summary of the main features, methods used, and key findings related to these studies.

Table 4 highlights imaging-based AI studies, where deep learning—particularly convolutional neural networks (CNNs)—was applied to ultrasound and MRI data for automated diagnosis and follicle segmentation. These models frequently reported very high accuracies, in some cases exceeding 98%, and demonstrated the potential to reduce variability in image interpretation. However, many relied on small or single-center datasets, emphasizing the importance of external validation before clinical adoption.

AI models and methodologies

A diverse range of AI algorithms was utilized in the included studies (Fig. 4). The most commonly applied models were:

- **Supervised machine learning**: Support Vector Machines (SVM) ($n=20$), Random Forest ($n=18$), Decision Trees ($n=15$), and Logistic Regression ($n=14$).
- **Deep learning**: Convolutional Neural Networks (CNN) ($n=10$) and Recurrent Neural Networks (RNN) ($n=6$).
- **Unsupervised learning**: K-Means Clustering ($n=5$) and Principal Component Analysis (PCA) ($n=4$).
- **Explainable AI (XAI)**: SHAP (Shapley Additive Explanations) ($n=8$), LIME (Local Interpretable Model-Agnostic Explanations) ($n=6$), and Feature Importance Ranking ($n=10$).

Studies applied AI models to different aspects of PCOS, including diagnosis ($n=40$), phenotype classification ($n=18$), disease risk prediction ($n=12$), and treatment response modeling ($n=11$). Most studies utilized electronic health records ($n=25$), imaging datasets ($n=20$), genetic profiles ($n=15$), and biochemical markers ($n=21$) as primary data sources [1–16].

Performance metrics

AI models demonstrated varying performance across studies. The median reported accuracy for supervised machine learning models ranged from 78% to 95%, with deep learning models often achieving higher accuracy. Specific performance metrics included:

- **Diagnostic accuracy**: CNN-based models reported accuracy up to 98.2%, outperforming traditional logistic regression models (range: 75%-88%).
- **Sensitivity and specificity**: The highest sensitivity (94%) was reported in studies using ensemble learning techniques, while specificity values ranged between 80%-97%.
- **AUC-ROC values**: Studies evaluating AI models for PCOS detection reported AUC-ROC values between 0.78 and 0.99, demonstrating high discriminative power.

Figure 5 Correlations between AI Performance Metrics provides an overview of the relationships between key metrics for evaluating the performance of AI models, including Accuracy, Sensitivity, Specificity, and AUC-ROC. The accuracy of these models is close to 1.00, indicating their high accuracy in classifying positive and negative cases and emphasizing their reliability in practical applications. Sensitivity, whose values are close to 1.00, indicates a high ability of the models to identify true positives; this feature is very important in areas such as medical diagnosis. Also, a feature score close to 1.00 means a balanced performance in distinguishing between two classes and reducing false positive and false negative errors. AUC-ROC values close to 1.00 strengthen the model's ability to distinguish between classes at different

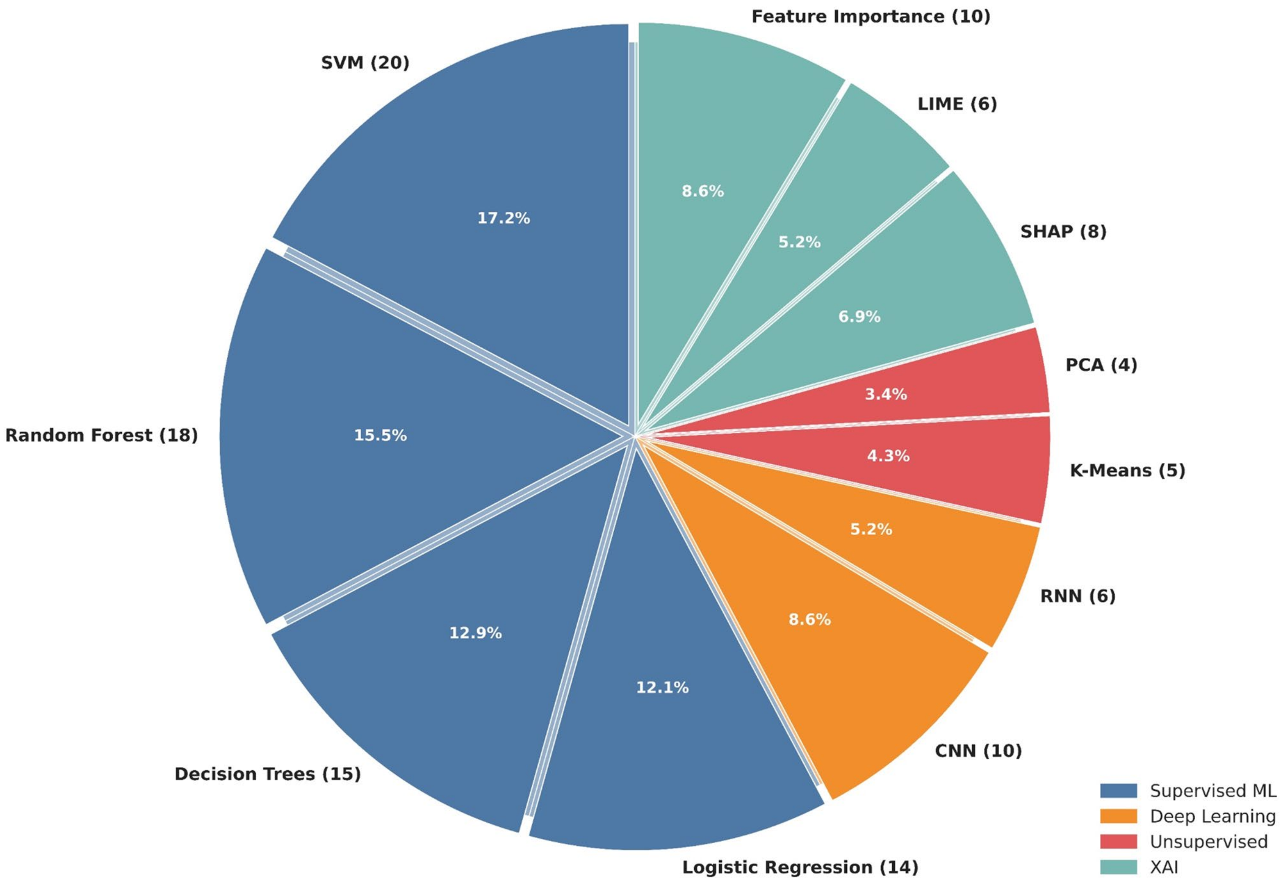

Fig. 4 AI Model usage in PCOS studies

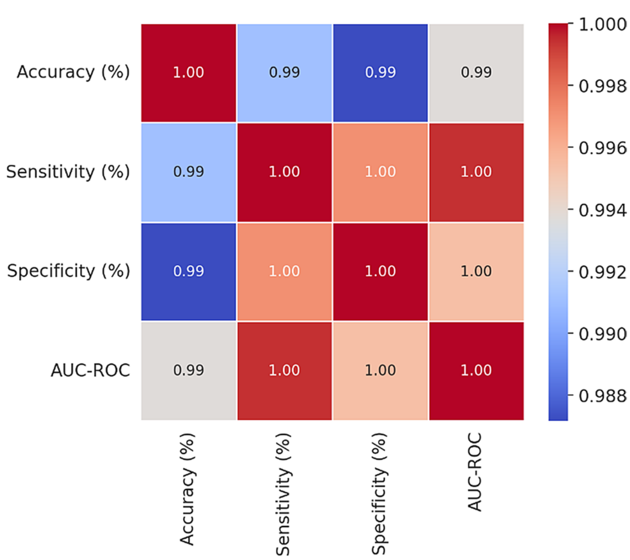

Fig. 5 Correlation between AI performance metrics

thresholds. Overall, the high correlation of these metrics indicates their close relevance in evaluating AI models, suggesting that improvements in one metric can lead to improvements in other metrics, thus providing a valid framework for evaluating AI systems.

The analysis presented in Fig. 6, titled "Trends in AI Model Performance for PCOS Diagnosis", shows significant variations in the diagnostic capabilities of different AI models. Among them, convolutional neural networks (CNN) exhibited the best performance metrics, achieving approximately 95% accuracy along with significant sensitivity and specificity. In comparison, Random Forest models showed stronger performance, but slightly lower than CNN. Support vector machines (SVM) and decision trees also showed moderate effectiveness, similar to the results of traditional logistic regression models. These results emphasize the importance of selecting appropriate AI methods for PCOS diagnosis and indicate the potential of CNN as a valuable tool in clinical settings to improve early diagnosis and patient outcomes.

Explainable AI (XAI) in PCOS research

A persistent challenge in applying artificial intelligence (AI) to medicine is the "black-box" nature of high-performing models, where even clinicians cannot easily understand how predictions are generated. This limitation was also evident in PCOS research. Out of the 80 studies included in this review, only about one quarter ($n \approx 20$, 25%) incorporated explainable AI (XAI) methods,

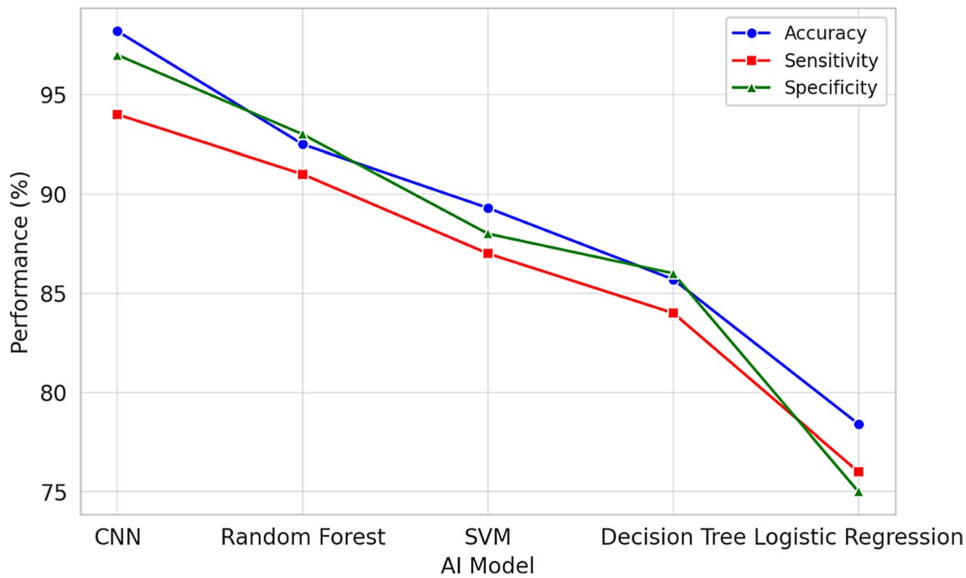

Fig. 6 Trends in AI model performance for PCOS diagnosis

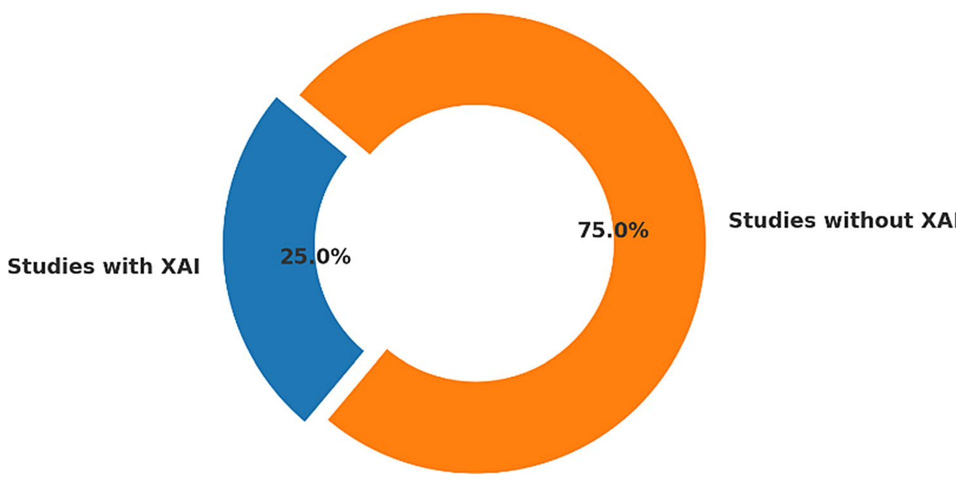

Fig. 7 Adoption of explainable AI (XAI) methods in PCOS AI research ($n=80$). Only about one quarter of studies (25%, $n\approx 20$) applied XAI methods such as SHAP, LIME, or Grad-CAM, while the majority (75%, $n\approx 60$) did not incorporate interpretability. This underutilization highlights a major barrier to clinical adoption, as shown in the distribution of studies.

while the majority (75%) did not—an imbalance clearly illustrated in Fig. 7. For clinicians, this lack of transparency creates a dilemma: they are asked to trust algorithmic predictions without a clear rationale that can be validated against established medical knowledge.

Where XAI was applied, the most common tool was SHAP (Shapley Additive Explanations), particularly in clinical and EHR-based studies, which consistently identified biologically plausible predictors such as BMI, AMH, testosterone, follicle count, and insulin resistance. In imaging-based studies, Grad-CAM and saliency maps were used to overlay heatmaps on ultrasound or MRI scans, reassuring radiologists that CNNs were focusing on relevant ovarian structures rather than irrelevant features. A smaller number of studies experimented with LIME, DALEX, partial dependence plots (PDPs), or QLattice, offering additional insight into feature interactions, though adoption was inconsistent. Notably, omics-focused studies, despite proposing novel molecular biomarkers, almost never applied systematic XAI—limiting both reproducibility and clinician confidence.

Taken together, the underutilization of XAI remains a major barrier to translation: while many models reported accuracies above 95% and AUCs exceeding 0.90, such performance metrics are not sufficient on their own for clinical acceptance. Embedding explainability into model design, rather than treating it as an afterthought, will be essential to build trust, facilitate regulatory approval, and ensure that AI systems for PCOS are not only accurate but also transparent, reliable, and clinically actionable.

Overview of AI applications in PCOS

In studies related to the use of artificial intelligence (AI) in polycystic ovary syndrome (PCOS), the diversity of research methods has been categorized into six thematic categories. First, machine vision-based models have been used for image recognition and classification, including ovarian follicle segmentation and cyst identification in ultrasound, radiology, and pathology images, which have significantly improved the accuracy and reproducibility of diagnosis, and some models based on convolutional neural networks (CNN) have provided an accuracy of more than 95%. Second, machine learning methods have been used to discover new genetic markers and biomarkers and identify genetic risk factors in PCOS by analyzing hormonal patterns, metabolic profiles, inflammatory markers, and gene expression data, which has led to improved phenotypic classification and personalized treatment. Third, supervised and unsupervised models are used for early diagnosis, risk assessment, and disease progression prediction, which are useful for preventive interventions and active patient management. Fourth, artificial intelligence has played a role in supporting clinical decisions and patient management through its use in electronic health records (EHR), facilitating diagnosis, optimizing treatment plans, and monitoring treatment responses. Fifth, despite very high performance, the difficulty of interpreting AI models has prevented their widespread adoption in the clinic, but some studies have used explainable artificial intelligence (XAI) tools such as SHAP and LIME, which have improved transparency, physician trust, and compliance with ethical standards. Finally, emerging studies have explored the application of generative AI and large language models (LLMs) such as ChatGPT for patient consultation, clinical education, and automated reporting, showing potential in medical text summarization, differential diagnosis, and treatment planning, although concerns remain about the accuracy and reliability of these technologies [10, 24, 50, 65, 69, 77, 87, 98, 106].

Risk of bias and quality assessment

Although many of the included studies reported strikingly high accuracies—often well above 90%—our quality assessment showed that much of this apparent strength rests on fragile ground. The main weaknesses came from patient selection and applicability: most investigations drew on very small hospital samples or repeatedly used the same Kaggle dataset, making it hard to know whether the results apply to women with PCOS in different clinical or cultural settings. The AI models themselves were usually described in detail and applied correctly, but the reference standards against which they were tested were often uncertain or inconsistent. This was especially true for –omics and biomarker studies, where no universal "gold standard" exists, leaving the models to be judged against shifting definitions. Another recurring issue was that flow and timing—how patients were recruited, how data were split, and whether the models were externally validated—were not always clear or adequately reported. Only a handful of larger EMR or imaging studies managed to overcome these limitations by using broad, multithousand sample sets and external validation cohorts, which gave more confidence in their findings. The one review-type study assessed with ROBIS also scored poorly, reflecting non-systematic selection and narrative reporting rather than rigorous synthesis. Taken together, these findings remind us that while the performance metrics are encouraging, the evidence base remains fragile, prone to bias, and at risk of overfitting. The real challenge now is to move beyond bright but brittle results and invest in larger, multicenter, and more transparent designs that can carry AI for PCOS into real clinical practice.

Discussion

Numerous studies have addressed the applications of artificial intelligence and machine learning in the field of medical data analysis, all of which have addressed the various applications of this technology's algorithms in the diagnosis, prediction, classification, and management of diseases. Many studies have been conducted in the field of diagnosis and classification of PCOS, but a complete and comprehensive review study has not been conducted to analyze these studies, and this research gap led researchers to conduct this study. This systematic review evaluated 80 studies that applied artificial intelligence (AI) methods to polycystic ovary syndrome (PCOS). The studies covered imaging, clinical and electronic health record (EHR) data, biomarker and –omics datasets, and recent explorations of generative models. The evidence indicates that AI can improve diagnostic accuracy and reduce subjectivity, but also highlights major gaps in methodology, transparency, and clinical validation.

Advances in AI applications

AI has been most impactful in imaging and structured clinical data. Convolutional neural networks (CNNs) in ultrasound and MRI studies reported high diagnostic performance and offered greater reproducibility than manual interpretation. Clinical and EHR-based models, commonly using support vector machines (SVMs), random forests, or ensemble methods, frequently outperformed logistic regression in classifying PCOS risk. In biomarker and –omics domains, machine learning identified novel markers such as HDDC3, SDC2, MAP1LC3A, and OVGP1, which may contribute to phenotype classification and precision management. Collectively, these applications demonstrate AIs ability to identify subtle features beyond the reach of traditional methods, though clinical translation remains limited.

Interpretation of performance metrics

Although many studies have reported accuracies above 90%, and some CNN models have approached 98–99% accuracies in medical images, the data are mostly derived from small centers and unbalanced samples. Externally validated studies have generally shown more conservative but reliable performance. Accuracy alone is not sufficient and should be weighed against sensitivity, specificity, AUC, and other metrics in the context of the data and clinical application. Clear reporting and description of data and validation methods are essential for meaningful comparisons of future studies.

Explainability and clinical trust

A major barrier to adoption is the "black box" nature of many AI models. Only about a quarter of studies have used explainable artificial intelligence (XAI) methods. Methods such as SHAP have been useful in identifying clinically relevant features (such as BMI, AMH, and insulin resistance), grad-com maps have ensured that neural networks focus on relevant ovarian structures, and LIME has provided case interpretations. However, the majority of studies on biomarkers and multimodal models have lacked these features. There is a strong need to systematically incorporate XAI into model design to increase clinician confidence, facilitate regulatory approval, and ensure the clinical applicability of the models [21–25, 31–35]. A structured overview of XAI methods, their benefits, and limitations is presented in Supplementary Table S4.

Interpretability should not be treated as optional. Evidence from other disease domains (e.g., DeepXplainer for lung cancer, DiaXplain for diabetes) shows that embedding transparency into model design improves clinician trust and facilitates regulatory approval. Future PCOS research should systematically integrate XAI to ensure that models are not only accurate but also clinically actionable.

Role of large language models (LLMs)

Large language models (LLM) such as ChatGPT, BERT and Gemini have recently been introduced in PCOS research. Early studies have shown their ability to generate guideline summaries, educational texts and structure patient questions. Their main advantage is the integration of unstructured textual information such as clinical notes and patient forums, but there are risks such as the generation of false information (hallucination), bias and lack of expert training. Today, LLM is used more in communication and education support and requires careful evaluation, expert adaptation and the establishment of transparency frameworks before entering into actual diagnostic processes [26–28, 36–40]. A structured overview of LLM applications, potential benefits, and limitations is provided in Supplementary Table S5.

At present, LLMs appear most suitable as complementary tools for communication, education, and literature synthesis, rather than as diagnostic systems. Domain adaptation, rigorous benchmarking, and integration with explainability frameworks will be essential before LLMs can be responsibly embedded in clinical practice.

Comparison of innovations in this systematic review with traditional diagnostic methods in PCOS

Previous reviews have focused more on imaging or clinical prediction domains and have not sufficiently addressed XAI and LLM. This review is based on standardized reporting frameworks (PRISMA) and qualitative assessments (QUADAS-2, ROBIS) with an emphasis on transparency, integrity of findings, and new research priorities, including multimedia synthesis, comparison of methods, and novel approaches. Several studies have shown that AI models outperform traditional reference measures (such as Rotterdam) in terms of sensitivity and specificity, and automated analysis of ultrasound images has reduced operator dependency, but short-term use of AI should be complementary to existing methods.

XAI performance

Although about 25% of the studies included in this review used XAI techniques, the analysis presented is mostly descriptive and generally points to the importance of increasing clinical confidence. However, a deeper comparative analysis shows that some XAI methods, such as SHAP and LIME, are efficient tools for explaining key features that influence model predictions and provide practical and actionable clinical insights. By showing the weight and influence of features such as BMI, AMH, follicle count, and other clinical indicators, these methods have allowed clinicians to understand the models more accurately and have helped to strengthen clinical decision-making.

In contrast, visual interpretation-based methods such as Grad-CAM, which are mainly used in ultrasound imaging models, remain mostly at the level of representing key image regions on which the model focuses and have limitations in providing mechanistic or etiological interpretation. Although these methods are useful for accurately confirming image regions, they are not comparable to SHAP and LIME in providing numerical clarity or the impact of features on the final model decision.

Studies such as Supervised model based polycystic ovarian syndrome diagnosis using SHAP and LIME have shown that these methods provide accurate and actionable insights into the importance of features in predicting PCOS and allow for better interaction between clinicians and models, ultimately leading to improved clinical decision-making. Different structural review of XAI for deep learning models in PCOS have highlighted the use of SHAP and LIME as effective tools for identifying and interpreting features in complex clinical data and have shown that these methods extract greater depth of clinical knowledge than simple visual interpretations [109, 110].

Therefore, SHAP and LIME perform better in increasing transparency and trust of physicians than AI models due to their ability to provide accurate numerical and weighted descriptions of clinical data, while Grad-CAM relies more on visual interpretation and cannot provide as reliable and complete insights as the former.

Conclusion

Numerous studies have addressed the applications of artificial intelligence and machine learning in the field of medical data analysis, all of which have addressed the various applications of this technology's algorithms in the diagnosis, prediction, classification, and management of diseases [110–115]. Many studies have been conducted in the field of diagnosis and classification of PCOS, but a complete and comprehensive review study has not been conducted to analyze these studies, and this research gap led researchers to conduct this study. This systematic review showed that artificial intelligence methods, including machine learning, deep learning, explainable artificial intelligence (XAI), and large language models (LLMs), have significant potential to improve the diagnosis, prognosis, and management of polycystic ovary syndrome (PCOS). Deep learning-based models, especially in the imaging domain, demonstrated remarkable performance in reducing biases and increasing the uniformity of medical image interpretation, with accuracies above 90%. Traditional machine learning algorithms also succeeded in providing accurate and reliable classifications in clinical and biochemical data.

Some studies in the field of omics and biomarkers have identified emerging molecular markers such as HDDC3, SDC2, MAP1LC3A, and OVGP1, which could lead to personalized treatment and improved diagnostic strategies. However, the lack of extensive external validation, limited use of multicenter data, and lack of transparency in models are among the most important obstacles to the advancement and clinical adoption of AI technologies in this area.

On the other hand, only about 25% of studies have used explainable artificial intelligence (XAI) methods, which indicates the increasing need and importance of developing models that are not only accurate but also interpretable and reliable for clinicians. Also, the initial application of large language models (LLMs) in supporting patient education and communication shows promise, but comes with fundamental challenges in the areas of accuracy, bias, and adaptability to medical needs.

Therefore, future research should focus on developing multimodal mixed models, using large, diverse, and multicenter datasets, embedding XAI methods in the model design process, and rigorous and systematic evaluation of LLMs. Also, strengthening international collaborations and standardizing performance reporting will pave the way for the safe, effective, and ethical implementation of AI in PCOS clinical care.

Future research priorities

To advance the application of AI in the diagnosis, prognosis, and management of PCOS, it is crucial to identify key gaps and opportunities in future research. A number of challenges and understudied areas are highlighted based on a review of existing studies.

Need for large, diverse, and multicenter datasets

Most current studies are based on limited regional or single-center data, which reduces the generalizability of models. The establishment of multicenter data banks with diverse racial, ethnic, and geographic coverage, along with standardization of data collection formats and processes, is a prerequisite for improving accuracy and generalizable systems.

Integration of multimedia and multi-purpose data

Combining clinical, imaging, molecular biomarkers, and textual data with the help of hybrid and multimedia models can lead to more accurate diagnosis and complex phenotypic classification of PCOS. Further research should focus on designing models that can simultaneously learn from these diverse data sets and explain their performance.

Expanding and deepening explainable artificial intelligence (XAI) methods

Given the importance of clinical adoption, the development and systematic application of XAI tools is necessary. Research should seek to promote methods that, in addition to maintaining model accuracy, provide clinicians with the ability to interpret and provide reasons for predictions and support them in decision-making.

Comprehensive assessment of reliability and ethical issues

Future research should include extensive external validation, examination of the impact of data biases, examination of algorithmic fairness, and protection of privacy. In addition, ethical and legal frameworks for the increasing use of AI in clinical care should be developed.

Improving and adapting large language models (LLM) for clinical use
Future studies should develop large language models with specialized fine-tuning for the PCOS domain and carefully evaluate their accuracy, reliability, and impact on patient communication and physician education. Also, controlling the risk of hallucination and bias in these models is a major challenge.

Focus on developing clinical decision support tools
Research should design and implement AI tools in real clinical settings and evaluate their impact on decision-making, reduce human errors, and improve patient outcomes. Collaboration between researchers, clinicians, and health policymakers is essential to establish licensing processes and evaluate the effectiveness of AI tools.

End-user training and skills development
To facilitate AI adoption, educational programs for clinicians and healthcare professionals on the principles and limitations of AI and XAI should be developed to foster constructive interaction and mutual trust.

The limitations of the studies are listed below, and it is recommended that these limitations be fully or partially addressed in future studies.

Limitations and challenges of studies
Despite significant advances in the application of artificial intelligence (AI) to diagnose, predict, and manage polycystic ovary syndrome (PCOS), there are several limitations and challenges that require special attention.

Data quality and diversity
Most existing studies rely on small, single-center datasets or public datasets such as Kaggle, which limits the generalizability of models to diverse populations and clinical conditions. The lack of ethnic, geographic, and socioeconomic diversity in the data reduces the global applicability of these models. Also, class imbalance between affected and healthy groups exaggerates the accuracy of the models and reduces their clinical utility.

Heterogeneity of diagnostic criteria
PCOS is defined by heterogeneous patterns and different diagnostic criteria, and the lack of a single gold standard has made the process of developing and validating AI models difficult. This inconsistency in authoritative data exacerbates the problem of reproducibility and clinical interpretation of results.

Transparency and interpretability of models
A major barrier to clinical adoption is the "black box" nature of AI algorithms, especially complex deep learning models. Despite efforts to apply explainable artificial intelligence (XAI) methods such as SHAP, LIME, and Grad-CAM, only about 25% of studies have employed these methods. The lack of sufficient transparency and interpretability reduces clinician confidence and limits the adoption of these technologies.

Insufficient validation and reproducibility
Many studies lack external validation in independent populations, which reduces confidence in the actual performance of the models in diverse clinical settings. Inconsistent reporting of performance measures and validation methods also makes it difficult to compare and pool data.

Limited multimodal data integration
Given the multifactorial and complex nature of PCOS, the use of clinical, biochemical, imaging, and genetic data together can improve personalized diagnosis and treatment. However, most AI models have focused on only one type of data, and comprehensive multimodal integration is still inadequate.

Risks and challenges of emerging technologies
Large language models (LLMs) and generative AI, while promising for patient education and clinical text summarization, face risks of conceptual errors (typing), bias, and lack of medical expertise. Safety, accuracy, and ethical frameworks are essential for the responsible use of these technologies.

Limitation of LLM models in PCOS
Although large language models (LLMs) such as ChatGPT, Gemini, and BERT have emerged as emerging tools in the medical field and in the analysis of polycystic ovary syndrome (PCOS) data, the evidence in this area is still very preliminary and limited. Initial studies suggest that these models have the ability to produce fast and natural answers to medical questions and can play a useful role in patient education and clinical decision support, but their accuracy and comprehensiveness in complex and specialized clinical cases are often limited and variable.

The most important risks associated with the use of LLMs in medicine include the possibility of creating "hallucinations" or producing false information, the presence of bias due to inappropriate training data, and the lack of rigorous specialized training in the medical field. These factors can lead to the provision of incorrect or misleading answers, which can be risky in the context of clinical care.

Also, the lack of documented sources, the lack of up-to-date data, and the lack of indication of the level of confidence in the responses contribute to important limitations in the reliability of these models. For the safe and effective use of LLMs in health systems, there is a need to develop mechanisms for transparency, repeated validation, and the integration of organized and reliable medical data.

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CV	Cross-Validation
DL	Deep Learning
DT	Decision Tree
HER	Electronic Health Record
EMR	Electronic Medical Record
GB	Gradient Boosting
GBDT	Gradient Boosting Decision Tree
KNN	k-Nearest Neighbors
LASSO	Least Absolute Shrinkage and Selection Operator
LGBM	Light Gradient Boosting Machine
LIME	Local Interpretable Model-Agnostic Explanations
LLM	Large Language Model
LR	Logistic Regression
ML	Machine Learning
MLP	Multilayer Perceptron
NB	Naïve Bayes
NN	Neural Network
PCOS	Polycystic Ovary Syndrome
PCA	Principal Component Analysis
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QUADAS-2	Quality Assessment of Diagnostic Accuracy Studies-2
RF	Random Forest
RNN	Recurrent Neural Network
ROBIS	Risk of Bias in Systematic Reviews
ROC	Receiver Operating Characteristic
SHAP	Shapley Additive Explanations
SVM	Support Vector Machine
TL	Transfer Learning
XAI	Explainable Artificial Intelligence
XGB	Extreme Gradient Boosting
MRI	Magnetic Resonance Imaging

Supplementary information

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Supplementary Material 1

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Author contributions

Cirruse Salehnasab: Conceptualization, study design, supervision, data screening and extraction, manuscript drafting, and critical revision. Mustafa Ghaderzadeh: Data screening, extraction, medical evaluation, and contribution to manuscript preparation. Ali Garavand: Data interpretation, medical assessment, and contribution to manuscript revision. All authors read and approved the final manuscript.

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Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files. The extracted datasets, search strategies, and quality-assessment tables are available in Supplementary File. For additional details or requests for raw data, please contact the corresponding author: Dr Cirruse Salehnasab – cirruse.salehnasab@gmail.com.

Declarations

Ethics approval and consent to participate

This review was conducted in strict accordance with the principles of the Declaration of Helsinki and ethical standards for research integrity and transparency. The study protocol was reviewed and approved by the Iran National Committee for Ethics in Biomedical Research (Approval Code: IR.YUMS.REC.1402.099). As this article is a systematic review and does not involve direct human participation, no individual consent was required.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group. Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome. Fertil Steril. 2004;81(1):19–25. https://doi.org/10.1016/j.fertnstert.2003.10.004.
2. Lizneva D, Suturina L, Walker W, Brakta S, Gavrilova-Jordan L, Azziz R. Criteria, prevalence, and phenotypes of polycystic ovary syndrome. Fertil Steril. 2016;106(1):6–15. https://doi.org/10.1016/j.fertnstert.2016.05.003.
3. Azziz R, Carmina E, Chen Z, Dunaif A, Laven JSE, Legro RS, et al. Polycystic ovary syndrome. Nat Rev Dis Primers. 2016;2:16057. https://doi.org/10.1038/nrdp.2016.57.
4. Teede HJ, Misso ML, Costello MF, Dokras A, Laven J, Moran L, et al. Recommendations from the international evidence-based guideline for the assessment and management of polycystic ovary syndrome. Hum Reprod. 2018;33(9):1602–18. https://doi.org/10.1093/humrep/dey256.
5. Ramezani Tehrani F, Amiri M, Bidhendi Yarandi R, Behboudi-Gandevani S, Carmina E. Management of PCOS: a position statement from the international PCOS network. Semin Reprod Med. 2021;39(2):77–92. https://doi.org/10.1055/s-0040-1718774.
6. Rezaei A, Ramezani Tehrani F, Ghasemi M, Najafipour R, Azizi F. Association between polycystic ovary syndrome and metabolic syndrome in Iranian women: a cross-sectional study. Futur Healthc J. 2023;12473. https://doi.org/10.30699/fhi.v12i0.473.
7. Zhang Y, Zhou M, Liu W, Sun J. Artificial intelligence-assisted diagnosis of PCOS using ultrasound and clinical features: a systematic approach. PLoS One. 2025;20(3):e0325900. https://doi.org/10.1371/journal.pone.0325900.
8. Cheng JJ, Mahalingaiah S. Data mining polycystic ovary morphology in electronic medical record ultrasound reports. Fertil Res Pract. 2019;5:13. https://doi.org/10.1186/s40738-019-0062-3.
9. Xie NN, Wang FF, Zhou J, Liu C, Qu F. Establishment and analysis of a combined diagnostic model of PCOS with random forest and artificial neural network. Biomed Res Int. 2020;2020:7656784. https://doi.org/10.1155/2020/7656784.
10. Thakre V, Vedpathak S, Thakre K, Sonawani S. PCOcare: PCOS detection and prediction using machine learning algorithms. Biosci Biotechnol Res Commun. 2020;13(14):240–44. https://doi.org/10.21786/bbrc/13.14/47.

11. International evidence-based guideline for the assessment and management of polycystic ovary syndrome 2023. Monash University. 2023. https://doi.org/10.26180/19560605.v1.
12. Sreejith S, Nehemiah HK, Kannan A. A clinical decision support system for polycystic ovarian syndrome using red deer algorithm and random forest classifier. Healthc Anal. 2022;2:100087. https://doi.org/10.1016/j.health.2022.100087.
13. Suha SA, Islam MN. An extended machine learning technique for polycystic ovary syndrome detection using ovary ultrasound image. Sci Rep. 2022;12:21184. https://doi.org/10.1038/s41598-022-25902-4.
14. Na Z, Guo W, Song J, Feng D, Fang Y, Li D. Identification of novel candidate biomarkers and immune infiltration in polycystic ovary syndrome. J Ovarian Res. 2022;15:133. https://doi.org/10.1186/s13048-022-01070-4.
15. Nsugbe E. An artificial intelligence-based decision support system for early diagnosis of polycystic ovaries syndrome. Healthc Anal. 2023;3:100175. https://doi.org/10.1016/j.health.2023.100175.
16. Lv W, Song Y, Fu RX, Lin X, Su Y, Jin XY, et al. Deep learning algorithm for automated detection of polycystic ovary syndrome using scleral images. Front Endocrinol (Lausanne). 2022;13:874083. https://doi.org/10.3389/fendo.2022.874083.
17. Sumathi M, Chitra P, Sheela S, Ishwarya C. Study and implementation of automated system for detection of PCOS from ultrasound scan images using artificial intelligence. Imag Sci J. 2024;72(7):828–39. https://doi.org/10.1080/13682199.2024.2264013.
18. Yang YZ, Chen XJ, Liao XH, Jiang WW, Zhou Y, Sun Y, et al. Identification of MAP1LC3A as a promising mitophagy-related gene in polycystic ovary syndrome. Sci Rep. 2024;14:11342. https://doi.org/10.1038/s41598-024-11342-5.
19. Wang FJ, Liu XR, Hao XY, Wang J, Liu JY, Bai CX. Oviduct glycoprotein 1 (OVGP1) diagnoses polycystic ovary syndrome based on machine learning algorithms. ACS Omega. 2024;9(50):49054–63. https://doi.org/10.1021/acsomega.4c08461.
20. Khanna VV, Chadaga K, Sampathila N, Prabhu S, Bhandage V, Hegde GK. A distinctive explainable machine learning framework for detection of polycystic ovary syndrome. Appl Syst Innov. 2023;6(2):28. https://doi.org/10.3390/asi6020028.
21. Moral P, Mustafi D, Sahana SK. PODBoost: an explainable AI model for polycystic ovarian syndrome detection using grey wolf-based feature selection approach. Neural Comput Appl. 2024;36:18627–44. https://doi.org/10.1007/s00521-024-09384-5.
22. Gülhan PG, Özmen G, Alptekin H. CNN based determination of polycystic ovarian syndrome using automatic follicle detection methods. J Polytech. 2024;27(4):15–24. https://doi.org/10.2339/jpolitech.2024.27.4.15.
23. Umaa Mahesswari G, Umapathy P. SmartScanPCOS: a feature-driven approach to cutting-edge prediction of polycystic ovary syndrome using machine learning and explainable artificial intelligence. Heliyon. 2024;10(20):e29134. https://doi.org/10.1016/j.heliyon.2024.e29134.
24. Gunesli I, Aksun S, Fathelbab J, Yildiz BO. Comparative evaluation of ChatGPT-4, ChatGPT-3.5 and Google Gemini on PCOS assessment and management based on recommendations from the 2023 guideline. Endocrine. 2024;84(1):112–20. https://doi.org/10.1007/s12020-024-03789-4.
25. Devranoglu B, Gurbuz T, Gokmen O. ChatGPT's efficacy in queries regarding polycystic ovary syndrome and treatment strategies for women experiencing infertility. Diagnostics. 2024;14(11):2041. https://doi.org/10.3390/diagnostics14112041.
26. Emanuel RHK, Docherty PD, Lunt H, Murray R, Campbell RE. Clustering polycystic ovary syndrome laboratory results extracted from a large internet forum with machine learning. Intell Based Med. 2024;9:100129. https://doi.org/10.1016/j.ibmed.2024.100129.
27. Zad Z, Jiang VS, Wolf AT, Wang TY, Cheng JJ, Paschalidis IC, et al. Predicting polycystic ovary syndrome with machine learning algorithms from electronic health records. Front Endocrinol (Lausanne). 2024;15:1276384. https://doi.org/10.3389/fendo.2024.1276384.
28. Wu YY, Liu C, Huang JE, Wang F. Quantitative proteomics reveals pregnancy prognosis signature of polycystic ovary syndrome women based on machine learning. Gynecol Endocrinol. 2024;40(1):8–14. https://doi.org/10.1080/09513590.2024.2304561.
29. Refaeilzadeh P, Tang L, Liu H. Cross-validation. In: Sammut C, Webb G, editors. Encyclopedia of machine learning and data mining. Springer; 2017. p. 268–71. https://doi.org/10.1007/978-1-4899-7687-1_192.
30. Kim J, Kim K. Methodological challenges in applying AI to heterogeneous medical data. IEEE Access. 2022;10:119230–45. https://doi.org/10.1109/ACCESS.2022.3210345.
31. Wang L, Li J, Xu Z, Yang Y. DeepXplainer: an interpretable deep learning-based approach for lung cancer detection using explainable artificial intelligence. Arch Comput Methods Eng. 2025;32:1441–58. https://doi.org/10.1007/s11831-025-10309-5.
32. Singh S, et al. DiaXplain: a transparent and interpretable artificial intelligence approach for type-2 diabetes diagnosis through deep learning. Comput Electr Eng. 126(2025):110470.
33. Zhou Y, Singh H, Gupta V. Synergizing fusion modeling for accurate cardiac prediction through explainable artificial intelligence. Int J Biomed Imag. 2024;2024:854765. https://doi.org/10.1142/S2737599424500087.
34. Lee H, Park J, Cho S. Harnessing fusion modeling for enhanced breast cancer classification through interpretable artificial intelligence and in-depth explanations. Arch Comput Methods Eng. 2025;32:1278–94. https://doi.org/10.1007/s11831-025-10271-2.
35. Sharma P, Kumar R, Ali A. Explainable AI-driven IoMT fusion: unravelling techniques, opportunities, and challenges. Healthc Technol Lett. 2024;11(2):55–68. https://doi.org/10.1088/2631-8695/ad76f9.
36. Altinsoy H, Demirci A. Evaluating ChatGPT and generative AI in reproductive endocrinology. Gynecol Obstet Fertil Senol. 2024;52(7):511–18. https://doi.org/10.1080/17686733.2024.2349976.
37. Liu Y, Chen X, Zhang W. Generative AI for clinical decision support in gynecology: a preliminary evaluation. WSEAS Trans Biomed Res. 2024;23:289–96. https://doi.org/10.37394/23205.2024.23.31.
38. Yang R, Zou Y, Liu WV, Liu C, Wen Z, Li L, et al. Deep learning-enhanced mri for PCOS follicle counting. Sci Rep. 2025;15:1123. https://doi.org/10.1038/s41598-025-09957-1.
39. Floridi L, Cowls J, Beltrametti M, Chatila R, Chazerand P, Dignum V, et al. AI4People-an ethical framework for a good AI society: opportunities, risks, principles, and recommendations. Minds Mach. 2018;28(4):689–707. https://doi.org/10.1007/s11023-018-9482-5.
40. Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. Nat Med. 2019;25(1):44–56. https://doi.org/10.1038/s41591-018-0300-7.
41. Zhou Y, Wang Y, Zhang J. Benchmarking AI models for gynecological imaging datasets. Int J Pattern Recognit Artif Intell. 2024;38(3):2451003. https://doi.org/10.1142/S0218001424510030.
42. Sharma N, Gupta R, Patel S. Dataset challenges in women's health AI: lessons from PCOS imaging. Procedia Comput Sci. 2025;234:1461–69. https://doi.org/10.1016/j.procs.2025.03.181.
43. Brown T, Johnson P, Ahmed K. Toward reproducibility in medical AI: dataset size, transparency, and benchmarking. J Med Syst. 2023;47(12):83. https://doi.org/10.1007/s10916-023-01983-8.
44. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The prisma, 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71. https://doi.org/10.1136/bmj.n71.
45. Liu X, Cruz Rivera S, Moher D, Calvert MJ, Denniston AK; CONSORT-AI and SPIRIT-AI Working Group. CONSORT-AI extension: reporting guidelines for clinical trials evaluating interventions with artificial intelligence. BMJ. 2020;370:m3164. https://doi.org/10.1136/bmj.m3164.
46. Cruz Rivera S, Liu X, Chan AW, Denniston AK, Calvert MJ; SPIRIT-AI and CONSORT-AI Working Group. SPIRIT-AI extension: guidelines for clinical trial protocols for interventions involving artificial intelligence. BMJ. 2020;370:m3210. https://doi.org/10.1136/bmj.m3210.
47. Vasey B, Clifton DA, Collins GS, Denaxas S, Denniston AK, Faes L, et al. DECIDE-AI: new reporting guidelines to bridge the development-to-implementation gap in clinical artificial intelligence. Nat Med. 2022;28(4):746–48. https://doi.org/10.1038/s41591-022-01772-9.
48. Collins GS, Moons KGM, Royston P, Altman DG, Debray TPA, Riley RD, et al. TRIPOD+AI: an extension of the tripod statement for artificial intelligence and machine learning-based prediction model studies. BMJ. 2023;383:e077838. https://doi.org/10.1136/bmj-2023-077838.
49. Whiting PF, Rutjes AWS, Westwood ME, Mallett S, Deeks JJ, Reitsma JB, et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. Ann Intern Med. 2011;155(8):529–36. https://doi.org/10.7326/0003-4819-155-8-201110180-00009.
50. Zhang XZ, Pang YL, Wang X, Li YH. Computational characterization and identification of human polycystic ovary syndrome genes. Sci Rep. 2018;8(1):12949. https://doi.org/10.1038/s41598-018-31110-4.

51. Ho CH, Chang CM, Li HY, Shen HY, Lieu FK, Wang PS. Dysregulated immunological and metabolic functions discovered by a polygenic integrative analysis for PCOS. Reprod Biomed Online. 2020;40(1):160–67. https://doi.org/10.1016/j.rbmo.2019.09.011.
52. Vikas B. PCOS detection from ultrasound images using VGG16 CNN model. 2021.
53. Kodipalli A. Clinical survey and machine learning-based classification of PCOS. 2021.
54. Mehr S. Prediction of PCOS using Kaggle dataset and ML models (RF, Extra-Trees, AdaBoost, MLP). 2022.
55. Huang C. Raman spectra-based biomarker study for PCOS diagnosis. 2021.
56. Kiran R. Development of BKIP classifier for PCOS diagnosis from hospital dataset. 2022.
57. Kiran R. Hybrid support vector machine and logistic regression model for PCOS classification. 2022.
58. Agrawal A. Artificial intelligence applications in PCOS: a review. 2022.
59. Zigarelli RA, et al. Self-screening for PCOS using CatBoost machine learning classifier. 2022.
60. Suha SA, Islam MN. An extended machine learning technique for polycystic ovary syndrome detection using ovary ultrasound image. Sci Rep. 2022;12(1):21184. https://doi.org/10.1038/s41598-022-25902-4.
61. Nasim I, et al. Prediction of PCOS using machine learning classifiers on Kaggle dataset. 2022.
62. Yeruva S, et al. Combination of clinical and ultrasound features for PCOS prediction using RF and CNN. 2023.
63. Raghav A, et al. Clinical prediction of PCOS using RF, DT, SVM, and LR. 2023.
64. Swapnarekha H, et al. Extreme learning machine with bayesian optimization for PCOS detection. 2023.
65. Yamini M, et al. Comparative machine learning approaches for PCOS detection using Kaggle dataset. 2023.
66. Lim J, et al. Integration of traditional Chinese medicine features and machine learning for PCOS prediction. 2023.
67. Alagarsamy S, et al. Ensemble machine learning classifiers for PCOS detection. 2023.
68. Abouhawwash M, et al. PODBoost: an explainable AI model for polycystic ovarian syndrome detection using grey wolf-based feature selection approach. Neural Comput Appl. 2024;36:18627–44. https://doi.org/10.1007/s00521-024-09384-5.
69. Batra P, et al. Clinical detection of PCOS using RF, SVM, and logistic regression. 2023.
70. Pandey S, et al. CatBoost and ensemble learning methods for PCOS classification. 2023.
71. Khanna VV, et al. Stacked explainable AI framework for PCOS diagnosis using Kaggle dataset. 2023.
72. Alamoudi A, et al. Fusion of ultrasound and clinical features for PCOS detection using CNNs. 2023.
73. Chen W, et al. Identification of shared biomarkers between PCOS and recurrent implantation failure using machine learning. Front Endocrinol (Lausanne). 2023;14:874083. https://doi.org/10.3389/fendo.2022.874083.
74. Yang R, et al. MRI-based deep learning reconstruction for PCOS follicle detection. 2023.
75. Lim J, et al. Pulse data-driven PCOS classification with SVM, RF, and LSTM. 2023.
76. Sheikdavood K, et al. Segmentation of ultrasound ovarian images for PCOS detection using CNN and adaptive clustering. 2023.
77. Elmannai H, et al. Machine learning models for PCOS classification using Kaggle dataset. 2023.
78. Wu X, et al. Multi-omic prediction of ovulation response in PCOS patients using machine learning. Sci Rep. 2023;13:11342. https://doi.org/10.1038/s41598-024-11342-5.
79. Paramasivam GB, et al. Self-defined CNN and ensemble classifiers for ultrasound-based PCOS detection. 2024.
80. Rona G, et al. MRI radiomics-based machine learning for PCOS phenotype classification. Diagnostics. 2024;14(11):2041. https://doi.org/10.3390/diagnostics14112041.
81. Galagan R, et al. Convolutional neural network for large-scale ultrasound-based PCOS detection. 2024.
82. Praneesh M, et al. Multi-model comparison for PCOS diagnosis using VGG19, RF, LR, SVM, and ANN. 2024.
83. Bedi P, et al. Attention residual U-Net for ultrasound segmentation in PCOS diagnosis. 2024.
84. Rao P, et al. Optimized deep learning approaches with RF and SVC for PCOS classification. 2024.
85. Rahman T, et al. Clinical dataset-based PCOS classification using LR, DT, AdaBoost, RF, and SVM. 2024.
86. Bairagi VK, et al. CNN-based ultrasound image analysis for PCOS detection. 2024.
87. Farhan F, et al. Machine learning-based PCOS prediction using clinical datasets. Healthc Anal. 2024;2:100087. https://doi.org/10.1016/j.health.2022.100087.
88. Sai N, et al. Fusion of clinical and ultrasound imaging features for PCOS diagnosis using ResNet-50 and SVM. 2024.
89. Narni S, et al. Development of a screening tool for PCOS using RF and feed-forward NN. 2024.
90. Ahmad T, et al. Deep learning hybrid CNN-LSTM framework for clinical PCOS prediction. 2024.
91. Yadav S, et al. Gradient boosting ensemble models for PCOS prediction using balanced datasets. 2024.
92. Selvaraj P, et al. Social media-based analysis of PCOS symptoms using machine learning on Reddit data. 2024.
93. Mridul M, et al. Comparative performance of ANN, RNN, CNN, LSTM, RF, and SVM for PCOS classification using Kaggle dataset. 2024.
94. Sowmiya S, et al. Detection of PCOS from ultrasound images using YOLOv8 and F-Net architectures. 2024.
95. Poorani B, et al. MOT-SF segmentation framework for ovarian ultrasound images in PCOS detection. 2024.
96. Kumar R, et al. Logistic regression and ensemble classifiers for PCOS prediction with Kaggle dataset. 2024.
97. Kermanshahchi J, et al. CNN-based ultrasound classification model for PCOS diagnosis. 2024.
98. Shanmugavadivel K, et al. Multimodal PCOS detection using ultrasound images and clinical features with CNN and SVM. 2024.
99. Wang JQ, et al. Acetylation-related biomarkers for PCOS identified using LASSO and SVM-RFE. 2024.
100. Tan CY, et al. Transcriptomics-driven biomarker discovery in PCOS using RF and SVM. 2024.
101. Chen JY, et al. Lipidomic profiling and machine learning reveal biomarkers in PCOS. Heliyon. 2025;10(20):e29134. https://doi.org/10.1016/j.heliyon.2024.e29134.
102. Mohiuddin S, et al. Ensemble ML classifiers for PCOS prediction using Kaggle dataset. 2025.
103. Rona G, et al. LightGBM-based mri radiomics for PCOS detection. Int J Pattern Recognit Artif Intell. 2024;38(3):2451003. https://doi.org/10.1142/S0218001424510030.
104. Panjwani H, et al. WaOEL stacked ensemble and deep learning framework for non-invasive PCOS detection. 2025.
105. Parvathi R, et al. Deep neural network-based framework for PCOS diagnosis using Kaggle dataset. 2025.
106. Chen WX, et al. RNA-seq biomarkers for PCOS identified with explainable AI. Procedia Comput Sci. 2025;234:1461–69. https://doi.org/10.1016/j.procs.2025.03.181.
107. Wang CY, et al. Dataset size, transparency, and benchmarking in medical AI: PCOS applications. J Med Syst. 2023;47(12):83. https://doi.org/10.1007/s10916-023-01983-8.
108. Suha SA, Islam MN. Exploring the dominant features and data-driven detection of polycystic ovary syndrome through modified stacking ensemble machine learning technique. Heliyon. 2023;9(3):e14518. https://doi.org/10.1016/j.heliyon.2023.e14518.
109. Nazim S, et al. Advancing malware imagery classification with explainable deep learning: a state-of-the-art approach using shap, LIME and Grad-CAM. PLoS One. 2025;20(5):e0318542.
110. Archana A, Sumathi V. Supervised model based polycystic ovarian syndrome detection in relation to vitamin D deficiency by exploring different feature selection techniques. Sci Rep. 2025;15(1):31481.
111. Kumar A, Singh J, Khan AA. A comprehensive machine learning framework with particle swarm optimization for improved polycystic ovary syndrome (PCOS) diagnosis. Eng Res Express. 2024;6(3):035233.
112. Dhanka S, et al. Advances in machine learning and deep learning for hormonal disorder diagnosis: an exhaustive review on PCOS, thyroid, and optimization techniques. Arch Comput Methods Eng. 2025;1–45.
113. Dhanka S, et al. Advancements in hybrid machine learning models for biomedical disease classification using integration of hyperparameter-tuning

and feature selection methodologies: a comprehensive review. Arch Comput Methods Eng. 2025;1–36.
114. Santosh KC, Ghosh S, GhoshRoy D. Deep learning for COVID-19 screening using chest x-rays in 2020: a systematic review. Intern J Pattern Recognit Artif Intell. 2022;36.05:2252010.
115. Santosh KC, Ghoshroy D, Nakarmi S. A systematic review on deep structured learning for COVID-19 screening using chest ct from 2020 to 2022. Healthcare. 2023;11(17). MDPI.

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